

Effective Web Interface Design and Development for Video Accessibility Challenges

CSC-895 Project Report

**A report on the various accomplishments and
challenges faced during the YouDescribe web
application project work.**

Submitted by

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Summer 2020

Effective Web Interface Design and Development for Video Accessibility Challenges

A project report submitted to the faculty of

San Francisco State University

in partial fulfillment of

the requirements for

The Degree of

Master of Science

in

Computer Science

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Summer 2020

CERTIFICATION OF APPROVAL

I certify that I have read the Website application for YouDescribe for SKI report by Poorva Rath, and that in my opinion her work meets the criteria for approving a project report submitted in partial fulfillment of the requirements for the degree: Master of Science in Computer Science.

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Abstract

This master project studied the development of diverse user friendly interfaces for Video Accessibility Research such as YouDescribeX application and Caption Rating Game. YouDescribe is making Youtube videos accessible to blind users. A very needed admin site for YouDescribe was created with effective interfaces for administrators to supervise and manage the data and community. Secondly, we are developing YouDescribeX with a Human-in-the-Loop Machine Learning (HILML) approach to provide users (describers and visually impaired people) a convenient way to utilize or interact Artificial Intelligence tools to extend the current YouDescribe. The upgrade to YouDescribe helped describers to lower the cognitive load and complete the tasks faster. Also interactive and fun webpages were designed for Caption Rating Game to evaluate how well a caption matches a provided image and effectively collect data set for YouDescribe. A good User Interface made it easier for the target audience to easily use the offered features. This report presents interface design and developments across multiple sites (YouDescribe admin, YouDescribeX, Caption Rating game) on Video Accessibility study.

I certify that the Abstract is a correct representation of the content of this thesis.

Chair, Thesis Committee

Date

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1. Introduction

This chapter covers the introduction of YouDescribe website. For the first section we discuss some background and motivation of the project. For the second section, we present the related work that has been done in the past.

1.1 Background and Motivation

Today videos are a source of information, education, entertainment and culture. Almost 5 billion videos are watched on youtube every day [18]. But these videos are not easily accessible for the visually impaired or blind due to heavy dependency on visual contents.

Video Inaccessibility for visually impaired and blind users:

- The WHO estimates that there are 285 million visually impaired people in the world, of which 246 million have low vision and 39 million are blind [19].
- The United Nations protocol states that a person with disabilities should be able to access services on the Internet and enjoy access to cultural material in accessible formats [21].
- The Web Content Accessibility Guidelines also recommends that audio descriptions should be provided for all online videos [21].
- Regardless of guidelines, 41% of the videos are incomprehensible without sound or captions, making it inaccessible to the visually impaired people[29].

YouDescribe Initiative:

YouDescribe is the collaborative project of SFSU and SKI (Smith-Kettlewell Eye Research Institute) in San Francisco, California[1]. It helps the sighted describers to describe the rapidly growing collection of Youtube videos for visually impaired users by providing audio descriptions since 2013. An additional soundtrack (audio

description) added by a describer helps the blind user to understand the text, illustrations, setting, gestures and facial expressions.

Below are some of the important and commonly used terms in YouDescribe:

- **Describers:** They are the sighted volunteers who are willing to convey the information, details, or particulars of videos in speech. These included professional describers who provided quick descriptions, volunteers and family members.
- **Viewers:** They are users who have low vision or are visually impaired, which includes fans who want to watch Youtube videos, blind university students or employees. It is essential at schools and workplaces to provide training for employees and classroom teacher's so that study material is accessible to blind students and employees.
- **Audio descriptions:** They are supplementary descriptions (voice over) tracks that provide visual information for blind viewers. Describers sync the recorded description, the original audio and the video.
- The **types of videos** available on YouDescribe are exactly the same which are accessible on Youtube. A Youtube video can be described and made available in YouDescribe upon request by a user. Or a describer can describe a video of their choice.
- **Wishlist:** A section that displays a set of videos for which audio description is requested by the viewers in the YouDescribe website. Describers can go through the wishlist and generate descriptions. Viewers can add votes for videos to increase the request of the video to be described.

- Describers have the ability to choose whether they want the descriptions to be **Inline** (a voice over) or **Extended** (option to pause the video in between for the description to be played) descriptions.
- **Waitlist** is a section where the user lands when the video needs some time to be preprocessed through Video Indexer. It will contain some preprocessed video from the wishlist section and sorted based on most wanted videos demanded by blind viewers that need describers to describe them according to categories.

Difficulties faced by Admin in YouDescribe:

YouDescribe works in the field of development for blind and visually impaired people. But to maintain the integrity of the product, an admin dashboard was required. The admin has to request the developer to perform actions (like remove inappropriate video or description, analyze the usage and ratings given by users, etc.) on any particular user, video or descriptions.

It is not accessible for admin to go over all the descriptions and videos. Admin is unable to restrict/ block inappropriate descriptions. Admin is unable to send feedback to the describers.

There is no statistical analysis of the data like the total number of data records(count of videos, descriptions and audiotracks), number of audiotracks per description, word cloud of audiotracks (most common words used while describing), category of videos and summary of time ranges was present in the system.

Difficulties faced by Describers in YouDescribe:

- YouDescribe is providing a human-only approach which requires efforts from humans to record their voices and sync with the video content.
- Inexperienced describers struggle to produce high quality descriptions.
- Some shy volunteers often don't publish their audio description as they are not happy with their recorded voice.
- The describers have to start over if the audio description (recorded) is not correct.
- Reduced number of describers due to lack of user friendly interface.

Possible Artificial Intelligence solutions to solve problems faced by describers:

- More features were required in the field of machine learning and artificial intelligence to solve the issues of describers.
- These features included automated scene segmentation, text generation of images, and merging scenes by video indexer API using machine learning.
- Assisting sighted users to create video descriptions by combining automation using deep learning with human input and editing (human in the loop approach).
- Further the saved text will be converted to automatic machine generated voice to reduce the load of new and shy volunteers.
- Describers can continuously save text and convert to speech scene by scene to overcome the problem of starting over to save the audio again and again.

Importance of Interactive User Interface:

- Implementing the machine intelligence systems makes the User Interface of the application more complex.
- Research has indicated that interactive user interface has increased the accessibility for users[23].
- A good UI helped the describers to completely access the features and tools provided in the application.
- A single word or visual representation (such as an icon) can describe and clearly communicate to the user its action. Thus makes it easier to understand[32].

1.2 Related Work

Providing descriptions for blind and visually impaired users:

- Video accessibility for blind or visually impaired users became crucial as online videos on websites such as YouTube became important in day to day lives[26].
- A significant drawback in video accessibility for blind and visually impaired users is the time and cost to produce video descriptions professionally.
- Professional video description typically takes a minimum two-week time and can cost up to hundreds or thousands of dollars for a video [26], depending on its length.

Some of the similar applications that work in providing the platform to make videos accessible for blind users are discussed below.

LiveDescribe:

- It was the first known computer-assisted video description platform [2] as shown in figure 1.1.
- It was designed for real-time descriptions of live television broadcasts and videos.
- Some of the important features of LiveDescribe were
 - Automatic detection of silent periods where a describer could choose to add a description.
 - Allow descriptions to go beyond the silent periods by automatically pausing the video to accommodate extended descriptions.

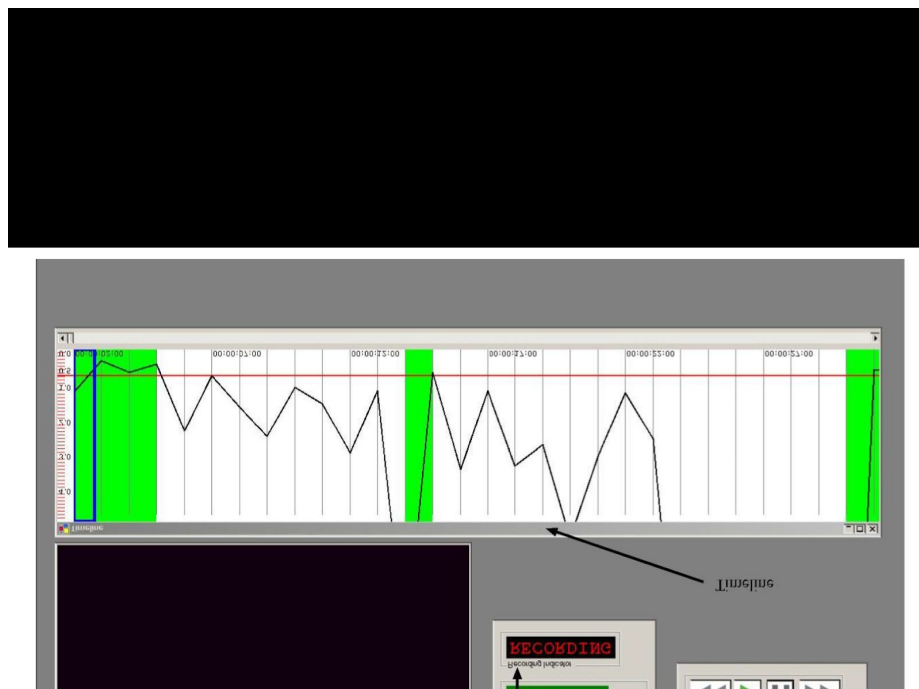


Figure 1.1. LiveDescribe user interface

Be My Eyes:

- It is an application that connects blind and visually impaired people with two million sighted volunteers for visual assistance through live and short video calls[3] as reflected in Figure 1.2.
- Authors revealed that Be My Eyes volunteers find the task and skills such as language and/or writing skills difficult and make them deterrents for providing video description[26].



Figure 1.2 Be My Eyes user interface

YouDescribe:

- It is a free platform that enables anyone to add an audio description to the videos uploaded on Youtube [1].

- YouDescribe works on humans-only approach which demand inputs from describers to record their voices and play along with the videos
- The application allows the user to control the generated scenes and merge similar scenes.

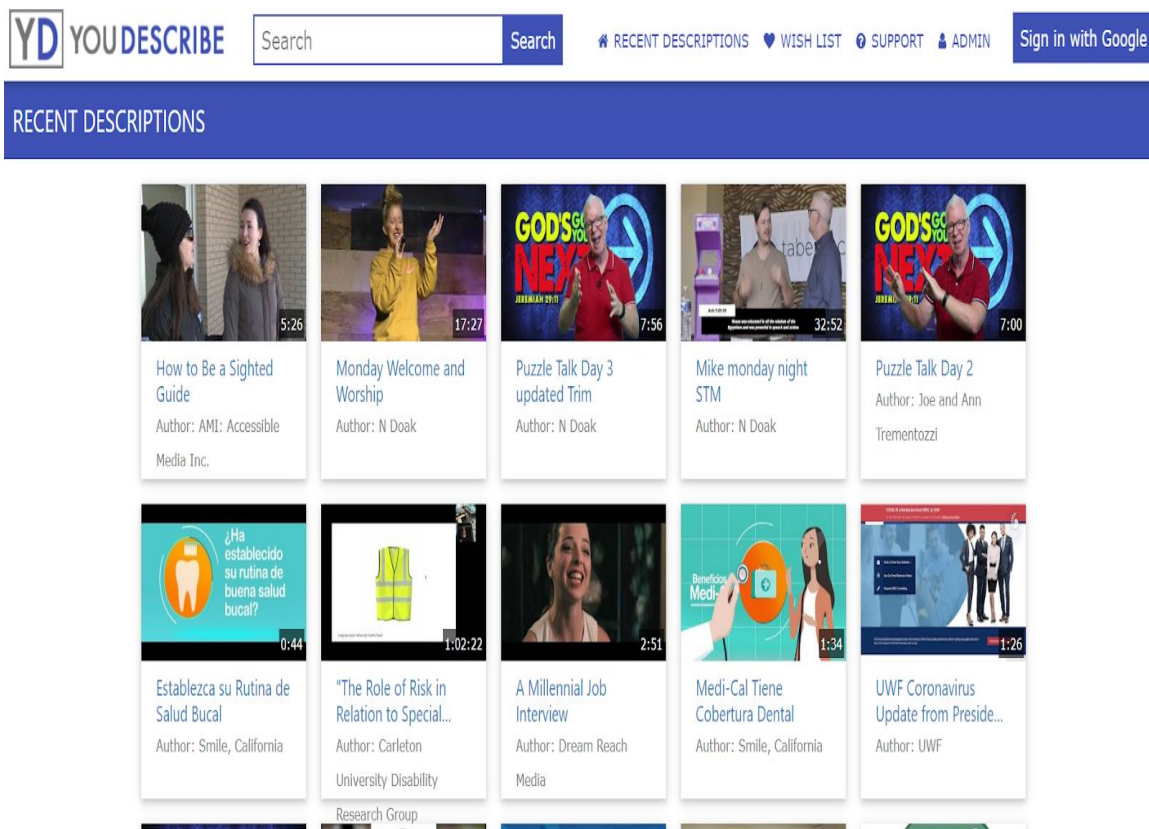


Figure 1.3. YouDescribe user interface

Problems faced by describers describing through above applications:

- The applications do not scale as each description is one-on-one and tailored to their experience.
- Informal feedback from volunteers in [26] showed that there are significant hurdles to create quality descriptions for videos in this way.

- Many of the volunteers do not have enough language knowledge or know how to provide **succinctly** informative and descriptive captions for videos that capture the correct visual elements.

Machine Learning advancement in improving the Video Descriptions:

The researchers illustrate the challenges of creating quality descriptions and develop a Human-in-the-Loop Machine Learning (HILML) approach to improving these captions[26]. They develop this approach to video description to reduce the cognitive load on the describer and simultaneously improve the overall quality of captions. The approach uses pre-generated captions to prompt the describer and to guide their own descriptions. 16 out of 22 describers felt the HILML was easier for them to use[26]. However, another survey question shows that there was only neutral belief that the text was accurate. A follow-up survey of blind users also favored the HILML descriptions over Human-only descriptions.

In the next section, we will discuss the requirement analysis of the applications.

2. Requirement Analysis

An overview of the project's functional requirement, use case diagrams and technical requirements is given in this section.

2.1 Functional Requirement

Below sub sections highlight the important functional requirements of the three applications.

2.1.1 Admin Section

To maintain the quality of description and decorum of the application an administrative panel was necessary for YouDescribe. Lack of an admin panel forced the admin had to request the developer to perform actions on any particular user or video. It was very inconvenient and stressful for developers as well as admin of the organization. Thus it was created to resolve issues faced due to lack of admin access.

Requirement ID	Requirement Statement	Requirement Details
2.1.1.1	Admin shall be able to login	• Admin will be able to login using the email id and password. • After successful login, the admin will be redirected to the homepage of the admin panel.

2.1.1.2	Admin shall be able to migrate between tabs.	• Admin will be able to choose between the tabs like description, videos and statistics. • App should be able to migrate from one page to another through tabs.
2.1.1.3	Admin shall be able to sort the videos or descriptions.	• Admin should be able to sort according to various options like Description Status, Descriptor, Video status etc. • App should be able to perform multiple sorts.
2.1.1.4	Admin shall be able to search the videos or descriptions.	• Admin should be able to search through Search-by-title, search-by-tag and search-by-describer. • App should be able to search through any keyword and display results.
2.1.1.5	Admin shall be able to click on the next/ previous button.	• Admin shall be able to redirect the video based on the search or sort videos or descriptions. • App should be able to right hand side on this detail page, to go to the next video and left hand side on this detail page, to go to the previous video.

2.1.1.6	Admin shall be able to publish/ unpublish video and give feedback to the describers.	• Admin have the access to publish/unpublish the video to youdescribe's home page. • Admin can send an email notification or feedback to the describer.
2.1.1.7	Admin shall be able to access the statistics module.	• Admin have the access to analyze records available in the database through the statistics model. • App will have a drop down option to select multiple records and diagrams to study the growth and dependency of the data.

2.1.2 YouDescribeX Section

To ease the jobs of describers using YouDescribe and make it convinient new features were to be added. YouDescribeX is a new version where describers of the YouDescribe website can go to the new interface to make use of the Artificial Intelligence to convert text to speech. Further select to add inline or extended descriptions.

Requirement ID	Requirement Statement	Requirement Details
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2.1.2.1	Describers shall be able to go to the new interface YouDescribeX.	• Descriptor will be able to go to the new interface simply by clicking on Button. • App should be able to display the button on each video when describers are describing the Youtube video. • If the video is available (processed) then the describers can start working on both Passes. • If the video is not available (not processed) then describers will land to the Waitlist section.
2.1.2.2	Describers shall be able to work on Waitlist section's videos.	• Describers will be asked to wait around some time till the video is processed. • Describers will be given the option to work on some preprocessed videos taken from wishlist and are not processed yet. • App will be able to notify the viewers by emailing the describer that their selected video is processed now. • App will be able to display the results of preprocessed video based on the most demanding Categories.

		• App will be able to remove and add more preprocessed videos, if any one describer describes that video. • If any video is selected, the app will be able to take the describer to two Passes.
2.1.2.3	Describers shall be able to work on Pass 1 (merging the scenes).	• Describers will be able to work on Pass 1 that is merging the scenes of the videos. • Describers will be able to view syncing of the videos and scenes. • App will be able to display all the scenes created by the Video Indexer API. • App will be able to display instructions to work on Passes. • App will be able to display each scene with their Scene number, Scene Starttime, Scene Endtime and button to merge scenes. • App will notify the user after merging of the scene is done.

		• Users can create notes based on each scene while watching the video.
2.1.2.4	Describers shall be able to work on Pass 2 (saving the descriptions).	• Describers will be able to work on Pass 2 after merging the scenes that are describing the Description and Text on Screen of each merged scene of the videos in words. • App will be able to display some auto generated text of Descriptions and users will be able to edit the same according to the scenes. • Describers will be able to save the Descriptions and the Text on Screen. • App will display saved notes of the user depending on the scenes. • App will be able to display each scene with their Scene number, Scene Starttime, Scene Endtime, Description textarea, TextOnScreen textarea and button to save the descriptions of the scenes. • App will notify the user after saving each scene.

2.1.2.5	Describers shall be able to work on Pass 2 (adding inline/extended descriptions) .	• Describers will be able to work on Pass 2 after saving the descriptions of each scene that are converting the text to speech in multiple sentences. • App will display the dialog timeline of the Youtube video and description timeline that is saved by the describer. • Describers will be able to change the description timeline to make it work either inline or extended description by radio button. • App will be able to merge the dialog timeline with description timeline to display the output as a described video.
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2.1.3 Caption Rating Game:

Caption Rating Game website is a fun and exciting gaming platform to identify the captions and pictures correctly. The game allows the user to rate the descriptions based on the contents of a given image. The idea is to find out the correct caption for the image, by providing multiple captions to a single image and the user needs to score the captions from 1 to 5 (5 being best). Further all the details are saved over the database and a score is calculated. After processing and analyzing the data, we will create a labeled training

dataset which is further used in the caption generation model. This will also be used to support AI generated video descriptions for blind and visually impaired users of the YouDescribe website.

Requirement ID	Requirement Statement	Requirement Details
2.1.3.1	User shall be able to Register	• New users will be able to register using user id, email id and password. • After successfully registering, users will be taken to the login page.
2.1.3.2	User shall be able to login	• Users will be able to login using the email id and password. • After successful login, the user will be displayed on the homepage where they can choose between training round or play.
2.1.3.3	User shall be able to choose between Training Round or Play.	• New users who are new to the game will be able to have a sneak peek of the game by clicking on the Training Round button. • Existing users who are aware of the game will be able to start playing the game where they have left by clicking on the Play button.

2.1.3.4	User shall learn about Instructions in the Training Round.	• New users will get instructions on how to play the game. • The app will show instructions only if users want to see it.
2.1.3.5	User shall be able to play around Training Round	• Users will have some training rounds to get an idea about the game. • The app will have image, caption and 1 to 5 radio buttons to capture user input. • Users will be able to rate the caption and will be able to get results on the basis of selection.
2.1.3.6	User shall be able to play	• The app will have 1 to 5 radio buttons to capture user input. • Users will be able to rate the caption and will be able to get results on the basis of selection. • Users will be able to see the scores. • Users will be able to go to the Dashboard.
2.1.3.7	User shall be able to see the scores Dashboard	• Users will be able to score each Round in Dashboard.

		• The app will display image, caption, score earned, consensus and total score.
2.1.3.8	User shall be able to see the Leaderboard	• Users will be able to score each player in Leaderboard. • The app will display player id and scores.

2.2 Use Case Diagram

Use case diagrams below explain the expected behavior of Users, and their implementation. These diagrams help to design a system from the end user's perspective. An effective technique to explain the system behavior as shown above in the Functional requirements section.

2.2.1 Admin section

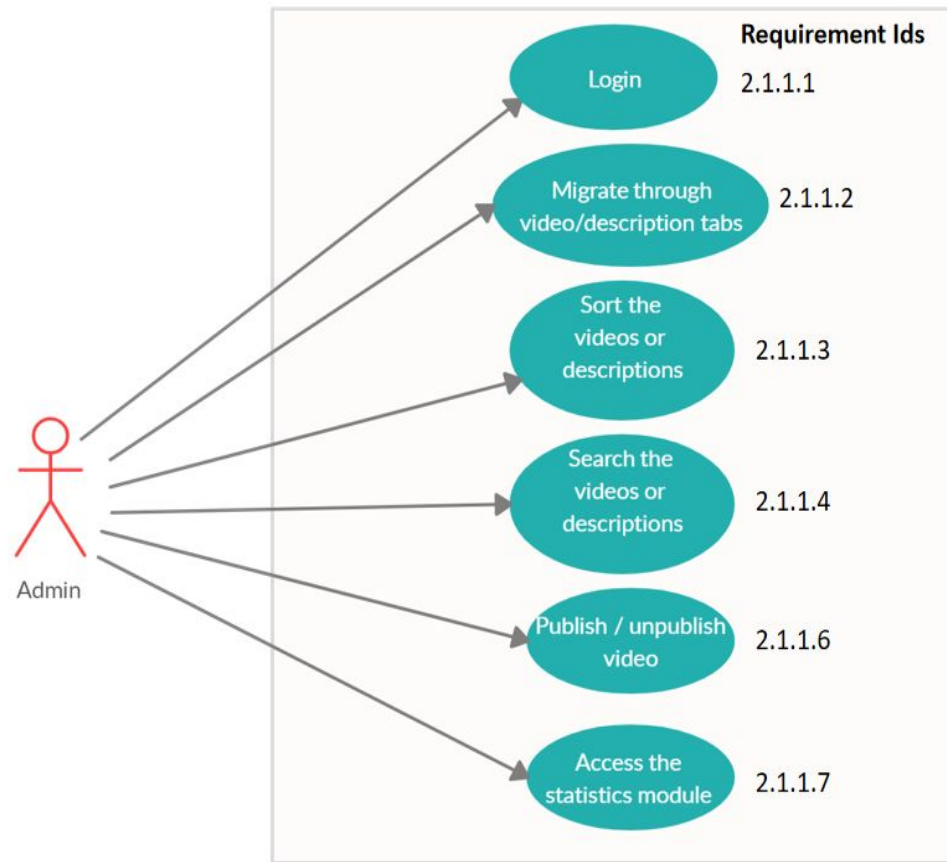


Figure 2.1: Use Case diagram for Admin Panel for YouDescribe application

2.2.2 YouDescribeX

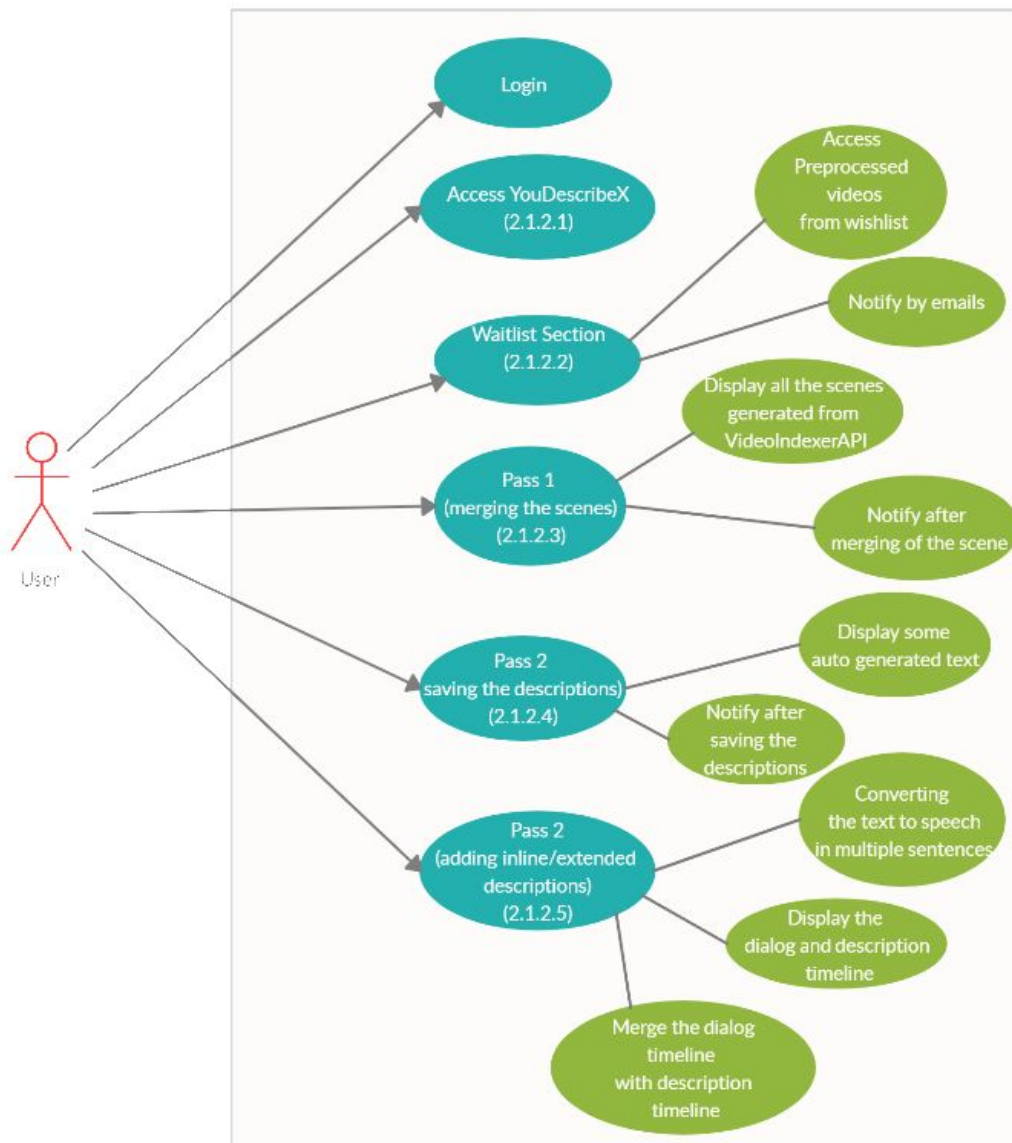


Figure 2.2: Use Case diagram for YouDescribeX Application

2.2.3 Caption Rating Game

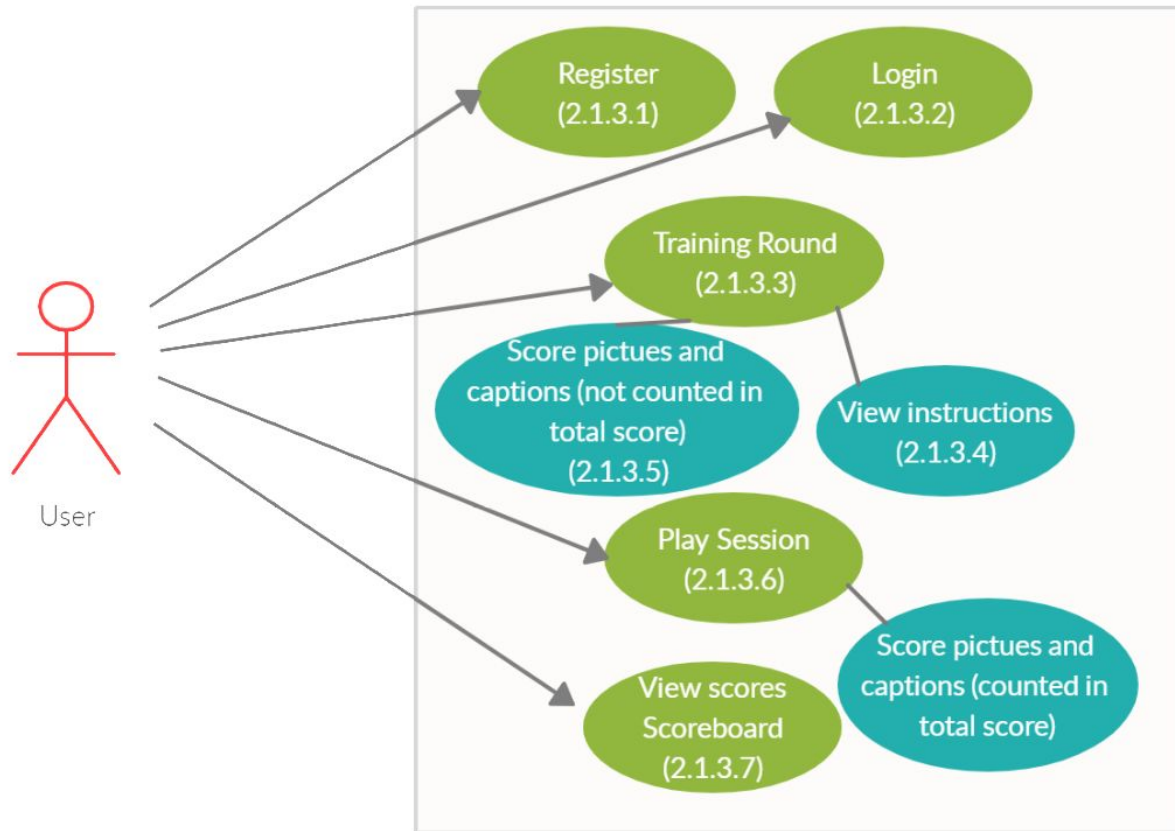


Figure 2.3: Use Case diagram for Caption Rating Game Application

2.3 Technical Requirements

After the idea of the application the objective was to study the various technologies that would be required to understand and implement the application. Some of the relevant technologies that would be required for the project were:

2.3.1 Youdescribe (Admin and YouDescribeX)

- Runtime Environment: NodeJS
- Database: MongoDB
- Backend Framework: ExpressJS
- Frontend Framework: React

2.3.2 Caption Rating Game

- Runtime Environment: NodeJS
- Database: MySQL
- Backend Framework: ExpressJS
- Frontend Framework: HTML, CSS

3. User Interface Designs

User Interface designs can make a very important and crucial impact on the application. To make the application user friendly and convenient we are focused on creating the UI designs for the three different applications and the frontend implementation. I created and designed the Admin panel, YouDescribeX and Caption Rating Game. Below are some mockup designs and screenshots. We created the UI mockups to receive feedback from Ms Charity from YouDescribe organisation and iterate through all the inputs provided for the admin panel and YouDescribeX.

3.1 Admin Panel

- 1) **Videos page:** It tells the admin what all videos are available on the server, illustrated in Figure 3.1. Here admin can click on each video and identify who all have described the videos. They can search through any username, video title, videoid or creation date. They can also sort the videos based on categories.

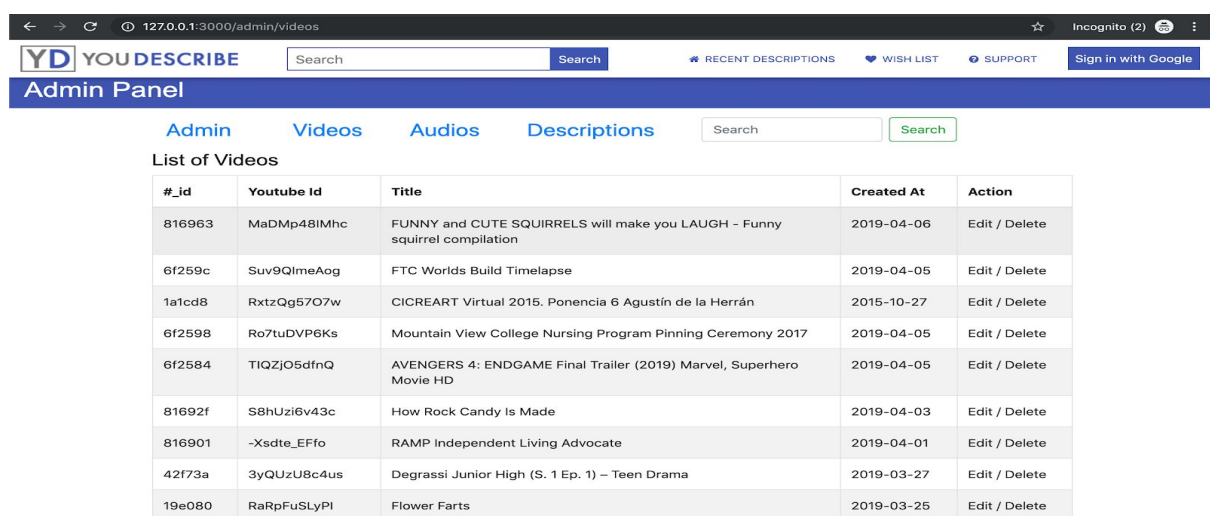
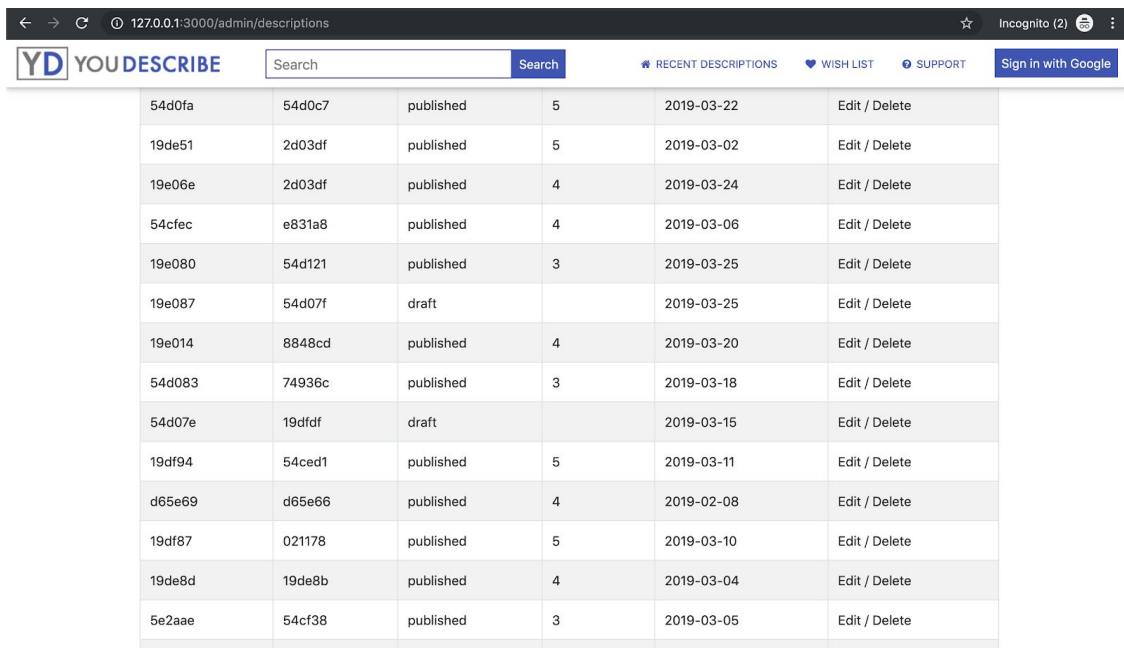


Figure 3.1: User interface mockup design for Videos page in Admin Section

2) **Description page:** It tells the admin what all descriptions are available on the server explained in Figure 3.2. Here admin can click on each description and identify who all have described the videos and whether the videos are published, unpublished or saved as draft. They can search through any username, video title, video-id or creation date. They can also sort the videos based on categories.



Video ID	User ID	Status	Count	Date	Actions
54d0fa	54d0c7	published	5	2019-03-22	Edit / Delete
19de51	2d03df	published	5	2019-03-02	Edit / Delete
19e06e	2d03df	published	4	2019-03-24	Edit / Delete
54cfec	e831a8	published	4	2019-03-06	Edit / Delete
19e080	54d121	published	3	2019-03-25	Edit / Delete
19e087	54d07f	draft		2019-03-25	Edit / Delete
19e014	8848cd	published	4	2019-03-20	Edit / Delete
54d083	74936c	published	3	2019-03-18	Edit / Delete
54d07e	19dfdf	draft		2019-03-15	Edit / Delete
19df94	54ced1	published	5	2019-03-11	Edit / Delete
d65e69	d65e66	published	4	2019-02-08	Edit / Delete
19df87	021178	published	5	2019-03-10	Edit / Delete
19de8d	19de8b	published	4	2019-03-04	Edit / Delete
5e2aae	54cf38	published	3	2019-03-05	Edit / Delete

Figure 3.2: User interface mockup design for Descriptions page

3) **Detailed description page:** This page displays when the admin clicks on any of the video from the video page. This is a detailed description which determines the videos, video ratings, and admin can publish/ unpublish the video directly from this page explained in Figure 3.3. The previous and next button works on the list that the user has selected on the video page. This means that if an admin has searched a user with name “Tom”, then all videos will display in the list/table format. Further if any video is selected from the list then previous and next buttons work on that list obtained.

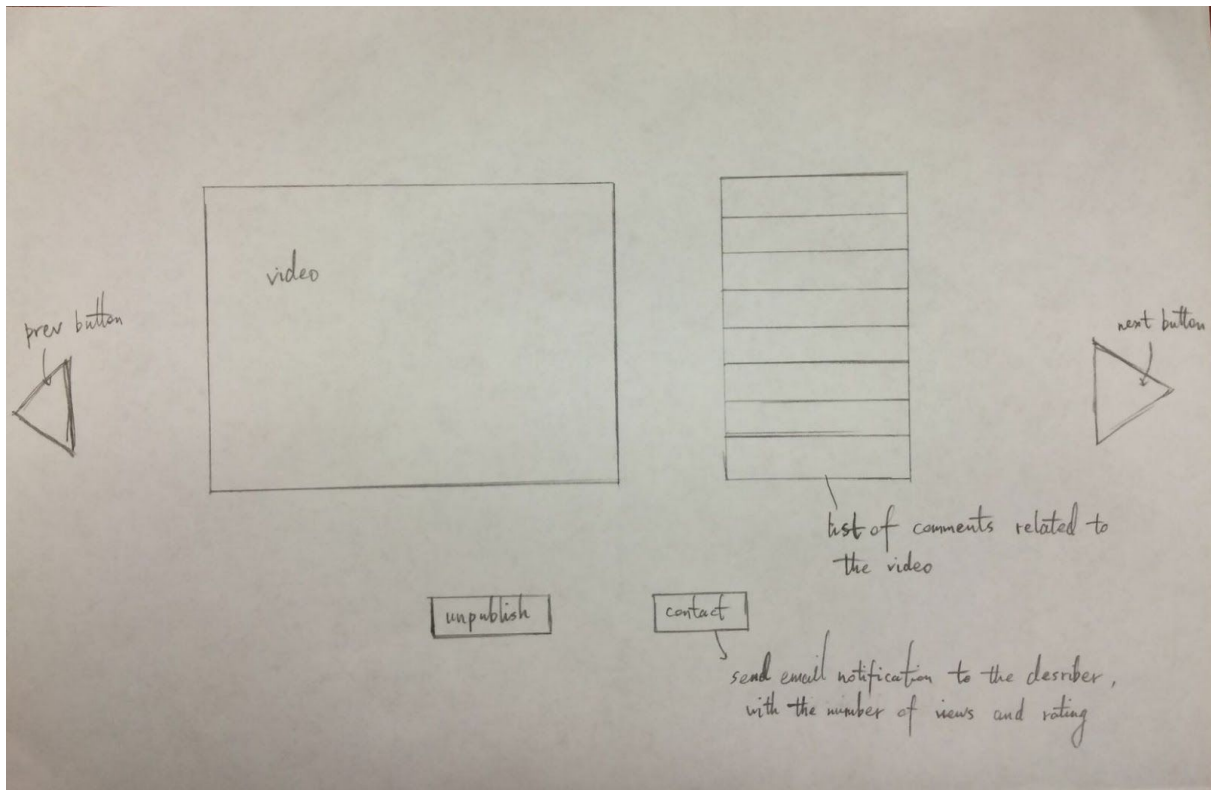


Figure 3.3: User interface mockup design for Detailed description page

With constant feedback and corrections in the UI, we came up with a new and improved interface which is explained in the Implementation and Results module.

3.2 YouDescribeX

- 1) **Access to YouDescribeX:** With the upgrade to YouDescribeX, users still can choose to use the old interface as recording their voice and adding the audio description to youtube content. Also they can try the new interface known as YouDescribeX illustrated in Figure 3.4. This is an artificially trained Human in the Loop model.

YouDescribeX redirects you to the webpage where the generation of automated scene segmentation, text generation of images, and text generation of text on screen using machine learning is done. The user is then able to edit the generated output, allowing human-machine collaboration to

produce high quality video descriptions while keeping a low barrier to entry for volunteer describers.

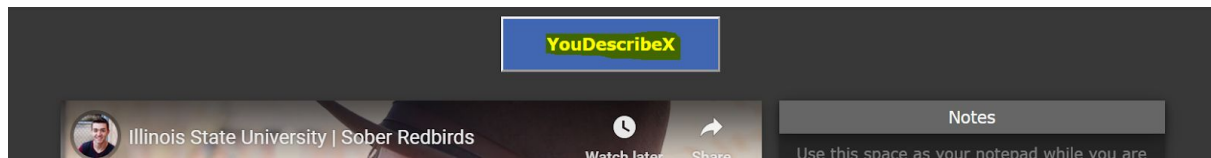


Figure 3.4: User interface mockup design to access YouDescribeX

- 2) **Waitlist Section:** The section where the video needs time to be preprocessed through Video Indexer API explained as shown in Figure 3.5. It also contains preprocessed videos from the wishlist section which are sorted based on the ratings given by blind viewers.

Thank You for choosing the new version of YOUDESCRIBE.

This video will be ready in 30 minutes. We will inform you via email once the video is ready. Below are some suggestions which are already processed. Do you like to describe them?

Suggestions for you.



Figure 3.5: User interface mockup design to access Waitlist section

- 3) **Pass 1:** The video is played first time to merge the scenes as shown in Figure 3.6.
 - a) It becomes easy for users to determine whether the current scene needs to be merged with the next scene.

- b) Users are allowed to merge unnecessary small segments done by the video indexer API.
- c) It is important to merge the scenes and only then add descriptions.
- d) Users will be able to create notes for each scene while watching the video.

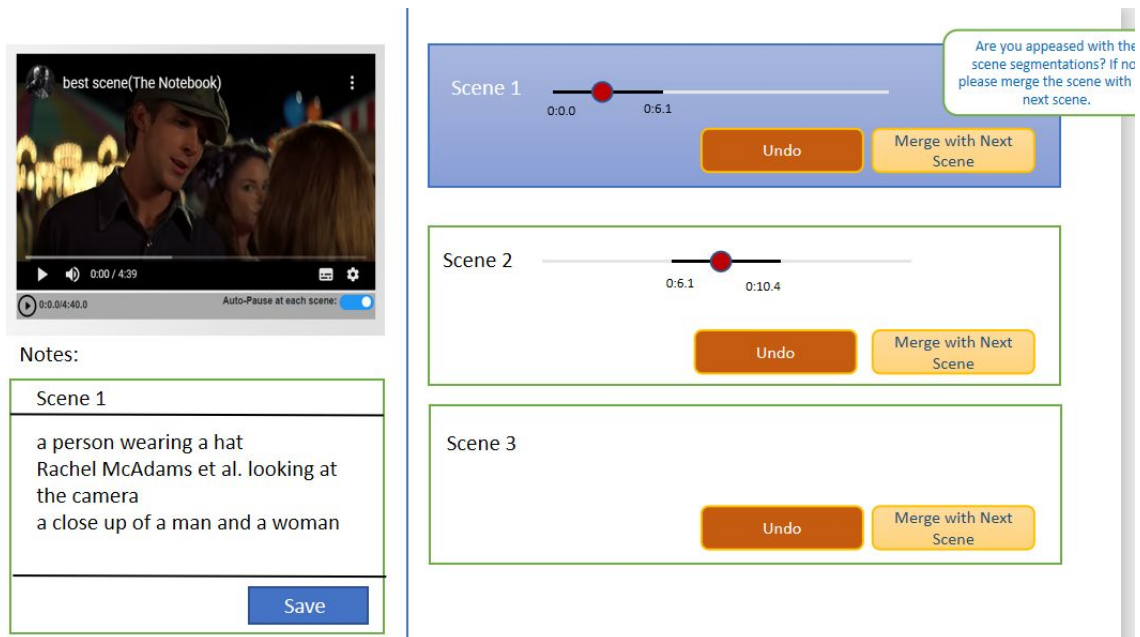


Figure 3.6: User interface mockup design of Pass 1 (merge scenes)

- 4) **Pass 2:** The video is played for second time:
 - a) After all the merging of scenes done by the user, the description can be edited.
 - b) The user will get text boxes for descriptions and text on screen to edit the text as explained in Figure 3.8 , the user will be able to edit based on the scenes. They will also be able to view saved notes.

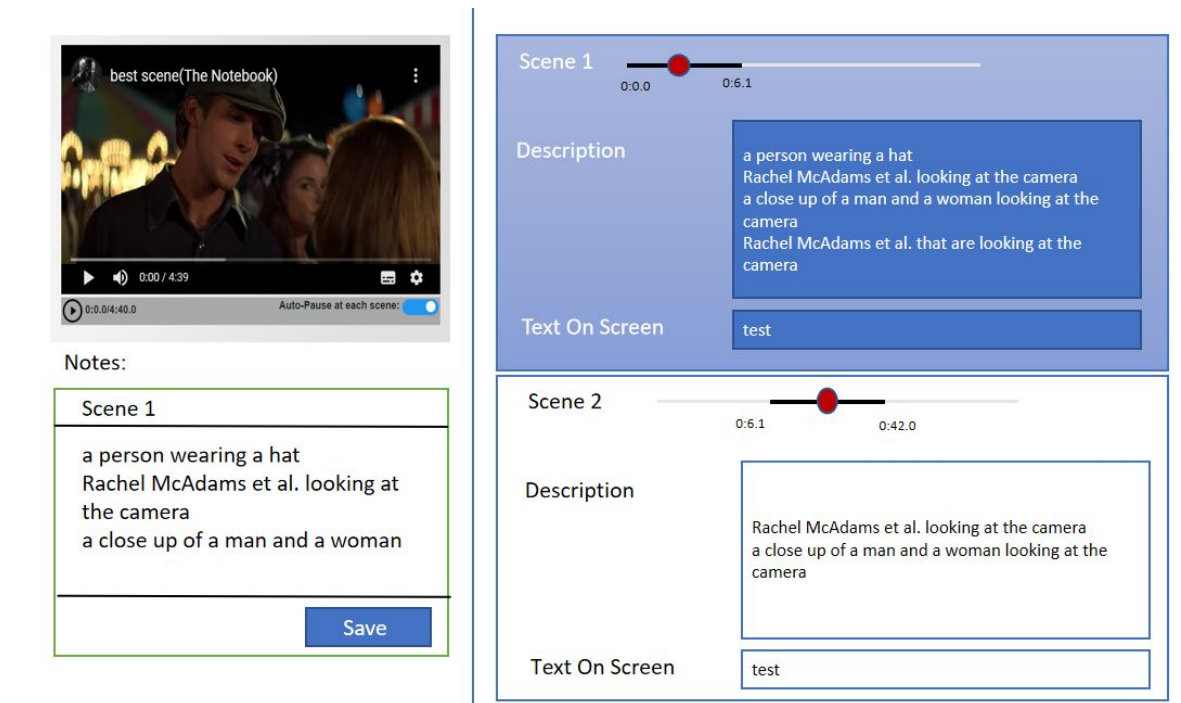


Figure 3.8: User interface mockup design of Pass 2

- c) Below in the Figure 3.9 explains the overall design to save the description and convert to description boxes.

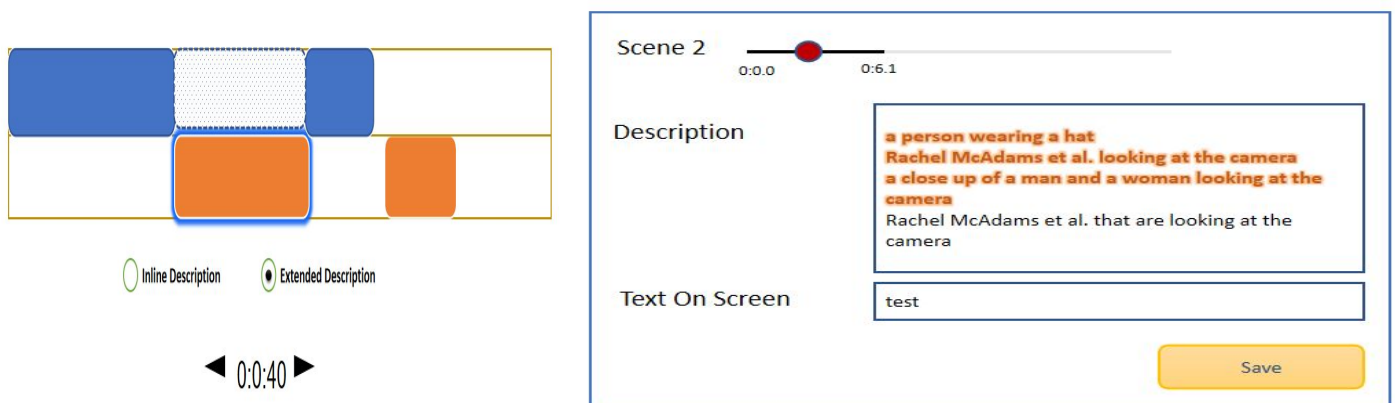


Figure 3.9: User interface mockup design of overall Pass 2

- d) When the user saves the description it will be represented as small boxes known as description timeline and maintained sentence by

sentence as shown in Figure 3.10. For each sentence there will be a separate audio file generated and will be shown as a box.

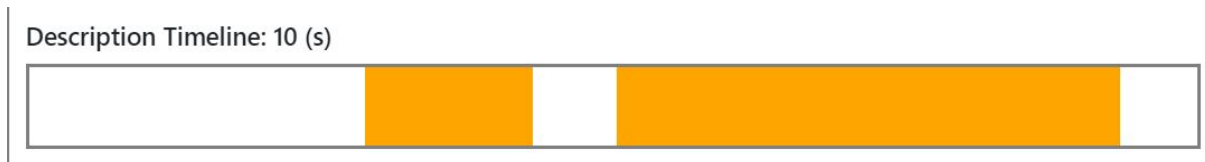


Figure 3.10: UI mockup design of rectangular boxes as per sentences.

- e) Users will be able to select the one description timeline box at a time as seen in Figure 3.11. The blue box refers to the dialog timeline and orange boxes refer to description timelines.



Figure 3.11: User interface mockup design of selection of one rectangular boxes at a time

- f) After selection, users can choose whether this description timeline box will be extended or inline, through a radio button as depicted in Figure 3.12.



Figure 3.12: User interface mockup design for selection of type of description based on selected rectangular boxes

- g) **Inline description:** Descriptions will be played along with the youtube video, and will be inserted with dialog timeline.
- i) Description is played without pausing the video. It can be mixed with the dialog timeline referred to Figure 3.13.



Figure 3.13: User interface mockup design of selection of Inline description merging audio with dialog.

- ii) Description can be played when the dialog timeline ends as shown in Figure 3.14.



Figure 3.14: User interface mockup design of selection of Inline description after dialog timeline in YouDescribeX

- h) **Extended description:** Descriptions will be played by pausing the video and extending the watch time as shown in Figure 3.15. Blue dotted box is the extended dialog timeline, where video is paused and description starts followed by the next video snippet.

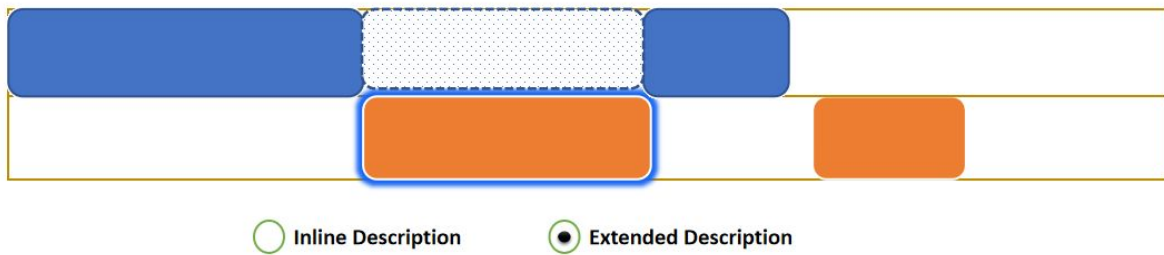


Figure 3.15: User interface mockup design of selection of Extended description pausing audio of dialog timeline

- i) User can also shift description timeline box left or right through buttons by some milliseconds or seconds as in Figure 3.16.

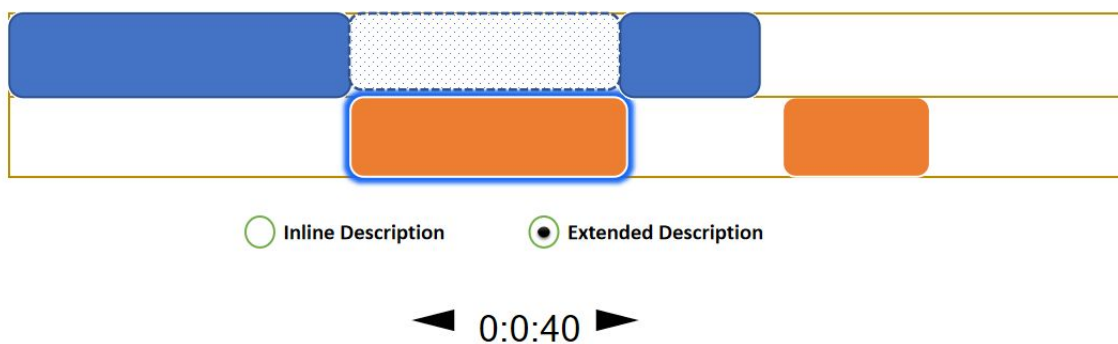


Figure 3.16: User interface mockup design of nudging timeline boxes

- j) While selecting the description timeline box it will highlight what scene and description of the scene is depicted in Figure 3.17.

The mockup shows a user interface for editing scene descriptions. It includes a timeline for 'Scene 2' with a red slider at 0:0.0 and 0:6.1. Below the timeline is a 'Description' section with a text box containing highlighted text: 'a person wearing a hat', 'Rachel McAdams et al. looking at the camera', and 'a close up of a man and a woman looking at the camera'. Below the description is a 'Text On Screen' section with a text box containing 'test'. A yellow 'Save' button is located at the bottom right.

Figure 3.17: User interface mockup design of highlighting text based on selection timeline boxes.

3.3 Caption Rating Game

- 1) **Home page of the game:** This is the landing page of the application, shown in Figure 3.18.

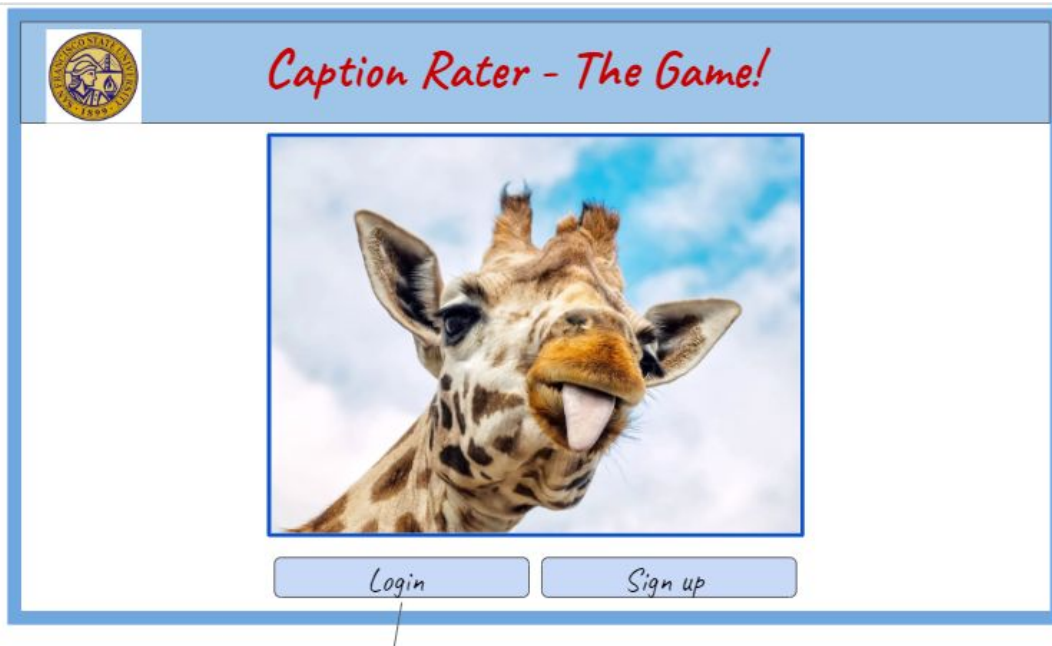


Figure 3.18: User interface mockup design of Home page

- 2) **Choose Home page:** Users can choose between the training round and play section reflected in Figure 3.19.

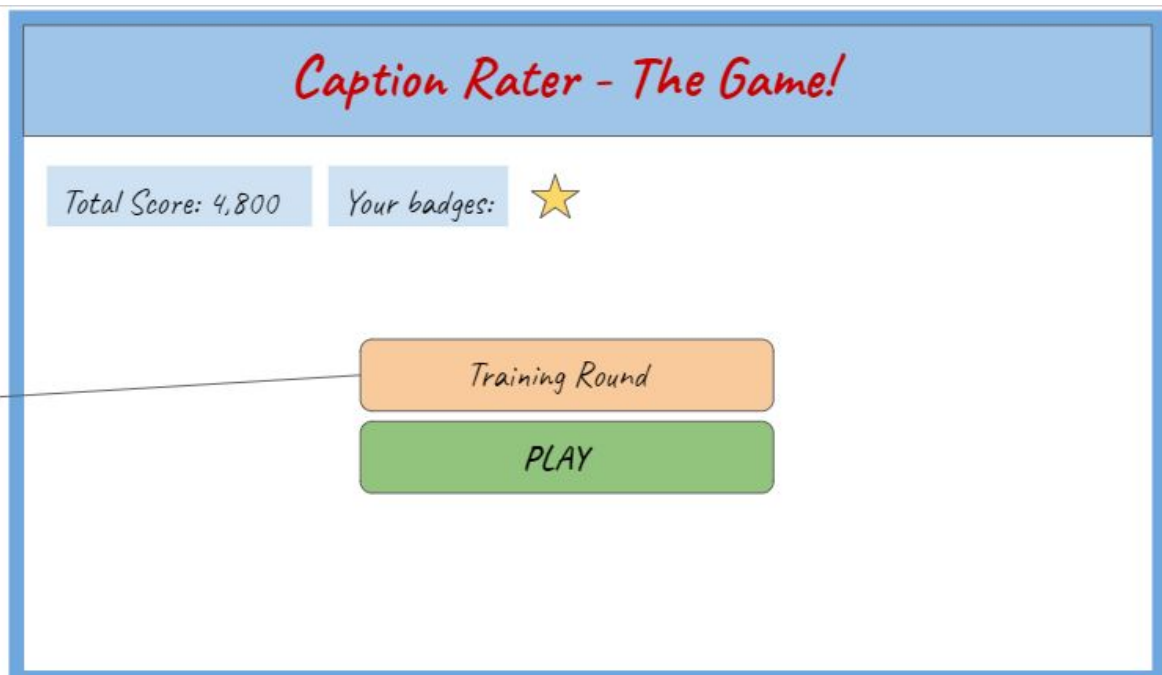


Figure 3.19: User interface mockup design of Choose Home page

- **Play Section:** User have to rate the given caption for an image ranging from 1 to 5 using radio buttons as described in Figure 3.20.

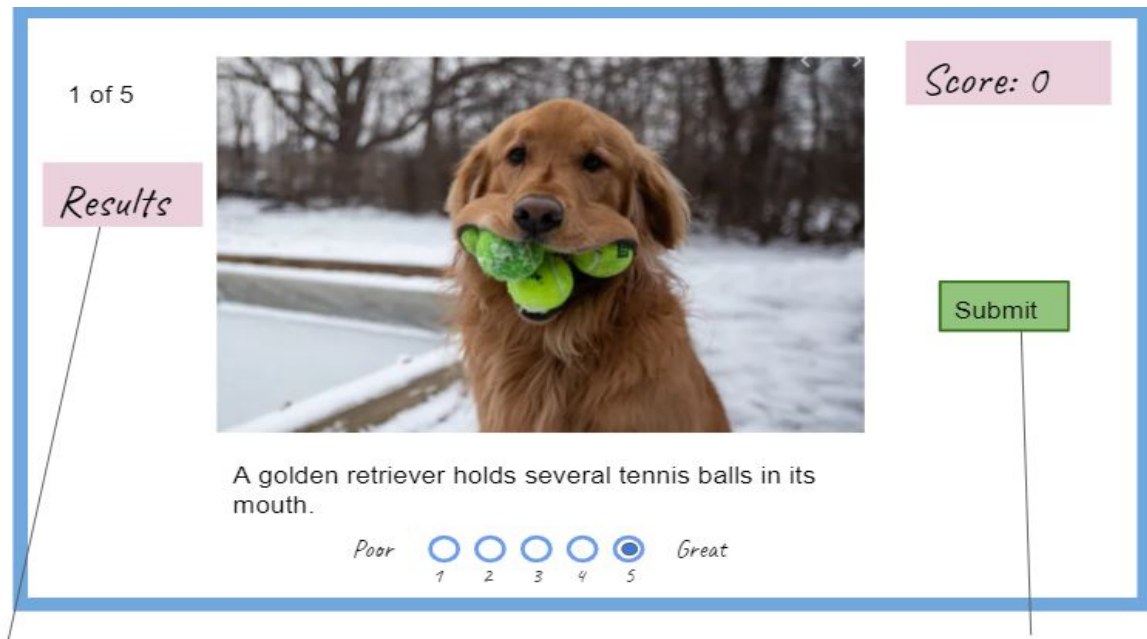


Figure 3.20: User interface mockup design of Play section

- **Dashboard:** This page displays the scores of the user for each image. Also consensus and points earned are calculated on the basis of the ratings given by the user as shown in Figure 3.21.

Score: 400		★ Total Score: 5,200		
Image	Caption	Your Rating	Consensus	Points earned
	A golden retriever holds several tennis balls in its mouth.	5	4.8	100
	Two people are driving in a car.	4	1	0
...	...	4	2	0
...	...	4	4	200
...	...	1	1.5	100
Exit		400		

Figure 3.21: User interface mockup design of Dashboard

4. Application Design and Architecture

In this section, the design and architecture of the application is described in detail. An application overview is given to provide a high level overview of the features provided by the application and to understand how the different technologies work together to constitute the app.

4.1 YouDescribeX Application Overview

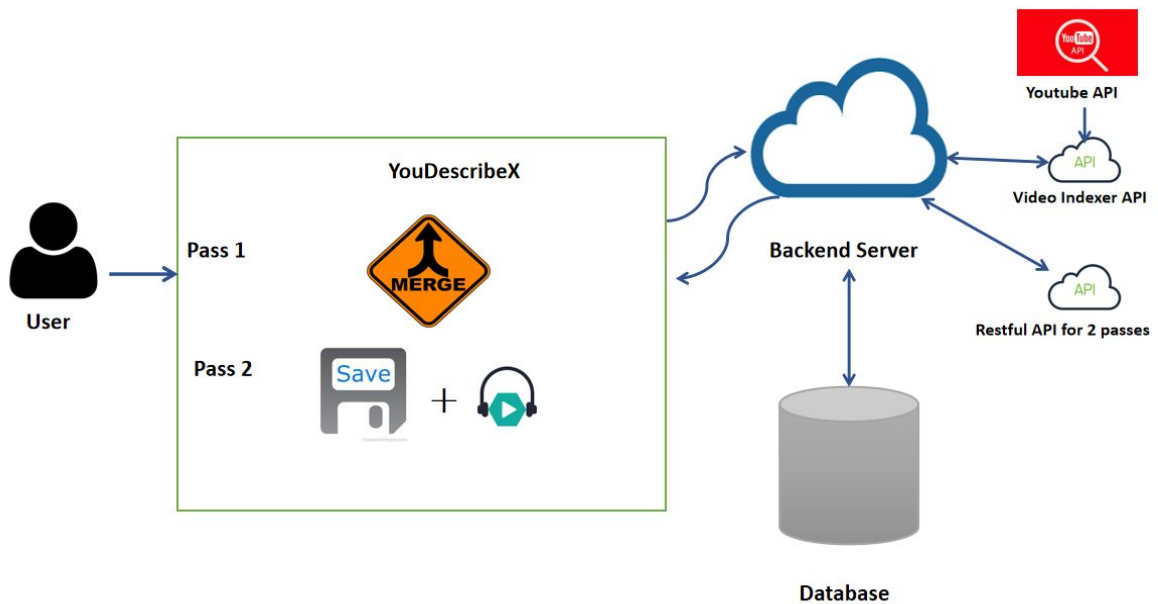


Figure 4.1: Overview of YouDescribeX

An overview of the YouDescribeX and the functionalities offered are shown in the above figure 4.1. The describer (user) can work on any youtube video. The describer will go through 2 Passes in the new version YouDescribeX. First the youtube video will be sent to VideoIndexer API to convert the video into small segments of scenes and to store auto generated descriptions and text on screens. VideoIndexer API will give all the information into JSON format that can be used in all the Passes. The first

Pass is Pass 1 where scenes generated from VideoIndexer will be displayed and the user has the option to merge scenes according to their conscience as some scenes are unnecessarily generated by VideoIndexer. For example 40 small scenes generated for a 2 minutes video is very time consuming. Further after merging, the user will be moved to the next Pass known as Pass 2. Here the user will get some auto generated descriptions generated from VideoIndexer. Users have the option to edit those Descriptions and Text on Screen. While working on Pass 2 users can view and convert Pass 2 saved descriptions into Text to Speech in the form of multiple audio files. After saving the description users can choose between inline or extended descriptions and play it along Youtube video , this is done scene by scene so users can listen to audio and video together. Changes can be done repeatedly working back and forth.

4.2 Architecture

4.2.1 YouDescribe

YouDescribe uses MERN (MongoDB, ExpressJS, ReactJS, NodeJS) stack for its development. The server resides on Amazon Web Services (AWS) as shown in Figure 4.2.

Web application: It consists of the React JS and the HTML, CSS, JavaScript code used to show the videos on the device. The videos are rendered as an iframe and shown on the webview. The components within the web application interact with the APIs.

MongoDB Database: The application uses a MongoDB database to store all the information in JSON format. The database can be accessed from the application at any time and is persistent.

Backend Server: The API services used by the application for video lookup can be accessed from within the application. The endpoints are secure and require internet access. AJAX calls can be made to the API endpoints to return JSON data which is then parsed and displayed by the application.

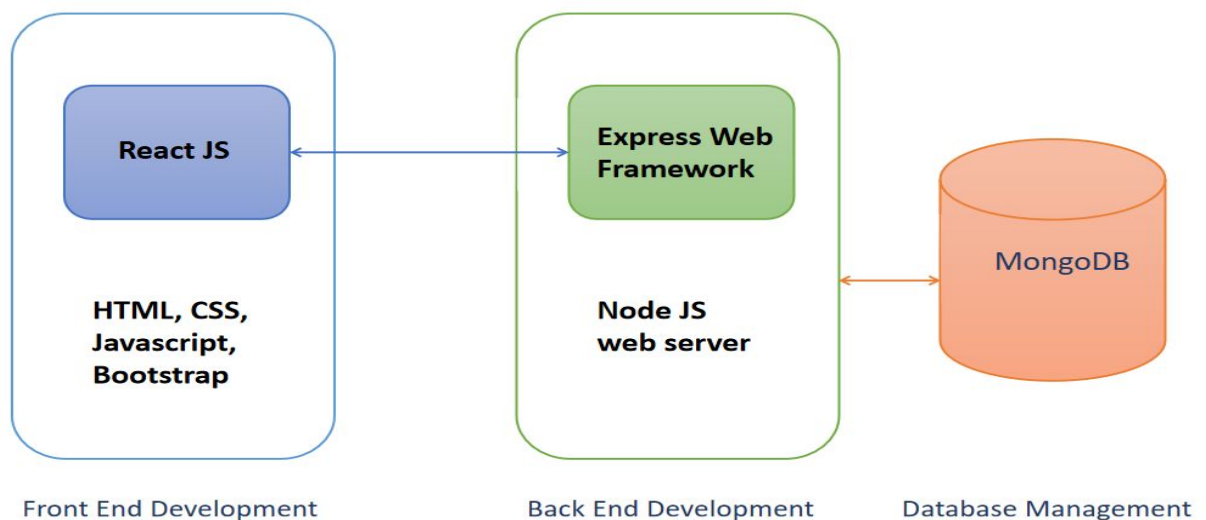


Figure 4.2: System Architecture for YouDescribe Website Application

4.2.2 Caption Rating Game

Caption Rating Game uses a standard Client-Server architecture as employed by many games. The server uses HTTP protocol. The figure below shows the client, server and MySQL database that is used. The server resides on Amazon Web Services (AWS) host (IP: 54.183.110.192) as depicted in Figure 4.3.

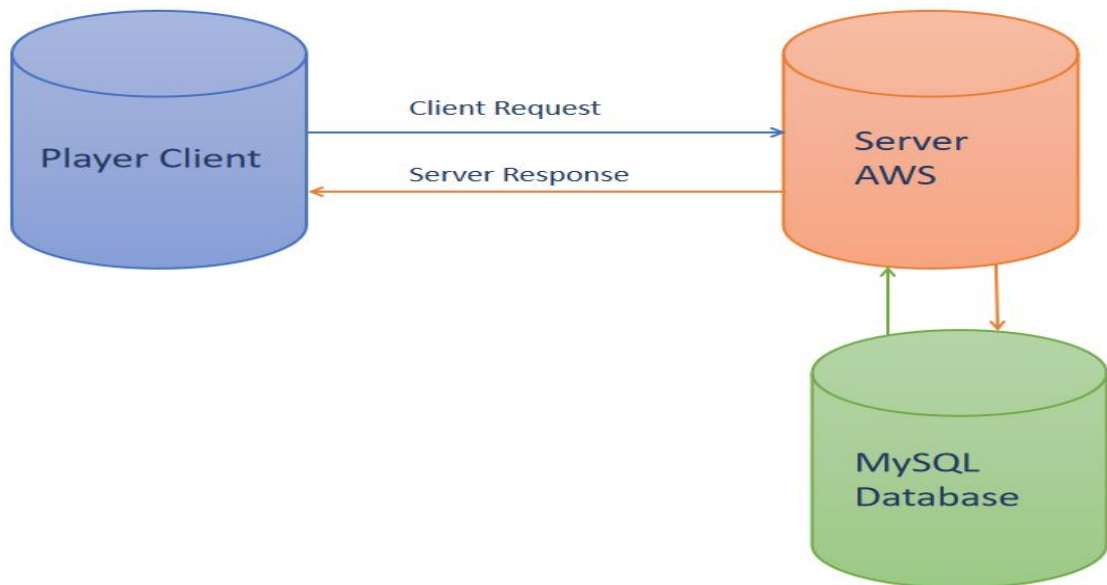


Figure 4.3: System Architecture for Caption Rating Game Application

4.3 Application Structure

4.3.1 Routes

Routes sequence	Routes explanation	URL
1	To retrieve specific video based on video id	http://dev.youdescribe.org:3001/api_version/videos/:id
2	To retrieve a specific user based on user id	http://dev.youdescribe.org:3001/api_version/users/:userId
3	To retrieve an array of videos on a specific page in wishlist	http://dev.youdescribe.org:3001/api_version/wishlist?page=:page

4.3.2 Views

Views sequence	Views	URL
1	Authoring-tool page	http://dev.youdescribe.org:3001/authoring-tool/:videoid
2	Search-by-title page	root_app/client/src/containers/video-page/component={SearchPage}?
3	Video page	http://dev.youdescribe.org:3001/video/:videoid
4	Wishlist page	http://dev.youdescribe.org:3001/wishlist
5	Credits page	http://dev.youdescribe.org:3001/credits
6	Contact page	http://dev.youdescribe.org:3001/contact
7	Support page	http://dev.youdescribe.org:3001/support
8	Support page (about)	http://dev.youdescribe.org:3001/support/about
9	Support page (describers)	http://dev.youdescribe.org:3001/support/describers
10	Support page (tutorial)	http://dev.youdescribe.org:3001/support/tutorial
11	Support page (viewers)	http://dev.youdescribe.org:3001/support/viewers
12	Unsupported-browser page	http://dev.youdescribe.org:3001/unsupported-browser

13	Error page	http://dev.youdescribe.org:3001/ *
----	------------	---

4.3.3 Database

i). db/connection.js

A config file is required for the connection to MongoDB, with all the configuration details in it. This file is not included in the repo, due to security reasons (sensitive credentials). Here is an example for the file of db/config.js.

- ```

config = {
 username: "your_username",
 password: "your_password",
 hostname: "127.0.0.1:27017",
 database: "testyoudescribe"
}

module.exports = config

```

### 4.3.4 Models

| Views sequence | Models                       | Models Explanation                                                                                                                     |
|----------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1              | root_api/models/audioClip.js | This is the model corresponding to the schema of audio clips. All the properties in this js are available in the table of audio_clips. |

|          |                                           |                                                                                                                                                                     |
|----------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2</b> | root_api/models/audioDescription.js       | This is the model corresponding to the schema of audio descriptions. All the properties in this js are available in the table of audio_descriptions.                |
| <b>3</b> | root_api/models/audioDescriptionRating.js | This is the model corresponding to the schema of audio description ratings. All the properties in this js are available in the table of audio_descriptions_ratings. |
| <b>4</b> | models/language.js                        | This is the model corresponding to the schema of languages. All the properties in this js are available in the table of languages.                                  |
| <b>5</b> | root_api/models/user.js                   | This is the model corresponding to the schema of users. Only the property of google_user_id in this js is available in the current table of users.                  |
| <b>6</b> | root_api/models/userVotes.js              | This is the model corresponding to the schema of user votes. All the properties in this .js are available in the table of users_votes.                              |
| <b>7</b> | root_api/models/video.js                  | This is the model corresponding to the schema of videos. All the properties in this .js are available in the table of videos.                                       |



|   |                             |                                                                                                                                     |
|---|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 8 | root_api/models/wishList.js | This is the model corresponding to the schema of wish list. All the properties in this .js are available in the table of wish list. |
|---|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

#### 4.3.5 Controllers

| Controllers sequence | Controllers                                               | Controllers Explanation                                                                                                                                                                                                                                                                                |
|----------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | root_api/controllers/audioClipsController.js              | <p>This is the controller to perform operations on audio clips.</p> <ul style="list-style-type: none"> <li>• addOne: Create an audio clip.</li> <li>• delOne: Delete an audio clip.</li> <li>• updateOne: Update an audio clip.</li> </ul>                                                             |
| 2                    | root_api/controllers/audioDescriptionsController.js       | <p>This is the controller to perform operations on audio descriptions.</p> <ul style="list-style-type: none"> <li>• updateAudioDescription: Update an audio description.</li> <li>• createOne: Create an audio description.</li> <li>• deleteAudioDescription: Delete an audio description.</li> </ul> |
| 3                    | root_api/controllers/audioDescriptionsRatingController.js | <p>This is the controller to perform operations on audio ratings.</p>                                                                                                                                                                                                                                  |

|   |                                             |                                                                                                                                                                                                                                                                                                                                                        |
|---|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                             | <ul style="list-style-type: none"> <li>• updateOverallRating: Update overall rating (helper function, not controller).</li> <li>• addOne: Create an audio rating (caller of updateOverallRating).</li> </ul>                                                                                                                                           |
| 4 | root_api/controllers/languagesController.js | <p>This is the controller to perform operations on languages.</p> <ul style="list-style-type: none"> <li>• getAll: Retrieve all languages.</li> </ul>                                                                                                                                                                                                  |
| 5 | root_api/controllers/usersController.js     | <p>This is the controller to perform operations on users.</p> <ul style="list-style-type: none"> <li>• getOne: Retrieve one user.</li> </ul>                                                                                                                                                                                                           |
| 6 | root_api/controllers/videosController.js    | <p>This is the controller to perform operations on videos.</p> <ul style="list-style-type: none"> <li>• getOne: Retrieve one video.</li> <li>• getVideoByUserId: Retrieve videos that are owned by a specific user.</li> <li>• getAll: Retrieve all videos.</li> <li>• search: Retrieve videos that include specific words in their titles.</li> </ul> |
| 7 | root_api/controllers/wishlistController.js  | <p>This is the controller to perform operations on videos.</p> <ul style="list-style-type: none"> <li>• addOne: Add a video to the wishlist.</li> </ul>                                                                                                                                                                                                |

|  |  |                                                                                                         |
|--|--|---------------------------------------------------------------------------------------------------------|
|  |  | <ul style="list-style-type: none"> <li>• <b>getOne:</b> Retrieve all videos in the wishlist.</li> </ul> |
|--|--|---------------------------------------------------------------------------------------------------------|

## 4.4 Database Schema

| Collection Name ^                          | Documents | Avg. Document Size | Total Document Size | Num. Indexes | Total Index Size | Properties                                                                            |
|--------------------------------------------|-----------|--------------------|---------------------|--------------|------------------|---------------------------------------------------------------------------------------|
| <a href="#">audio_clips</a>                | 5,651     | 364.3 B            | 2.1 MB              | 1            | 122.9 KB         |    |
| <a href="#">audio_descriptions</a>         | 606       | 881.2 B            | 534.0 KB            | 1            | 28.7 KB          |    |
| <a href="#">audio_descriptions_ratings</a> | 358       | 156.3 B            | 55.9 KB             | 1            | 24.6 KB          |    |
| <a href="#">countries</a>                  | 195       | 68.7 B             | 13.4 KB             | 1            | 20.5 KB          |    |
| <a href="#">languages</a>                  | 124       | 61.3 B             | 7.6 KB              | 1            | 20.5 KB          |    |
| <a href="#">reviews</a>                    | 0         | -                  | 0.0 B               | 2            | 12.3 KB          |    |
| <a href="#">users</a>                      | 2,420     | 58.0 B             | 140.3 KB            | 1            | 57.3 KB          |    |
| <a href="#">users_votes</a>                | 66        | 117.0 B            | 7.7 KB              | 1            | 20.5 KB          |  |
| <a href="#">videos</a>                     | 500       | 788.8 B            | 394.4 KB            | 2            | 98.3 KB          |  |
| <a href="#">wish_list</a>                  | 2,647     | 129.1 B            | 341.8 KB            | 1            | 65.5 KB          |  |

*Figure 4.4: Database Schema for the YouDescribe website.*

**4.4.1 audio\_clips:** This collection is used to save audio clips for each scene for the Youtube videos as shown in Figure 4.5.

```

_id: ObjectId("58e7cf4946e13dfd851a349b")
audio_description: ObjectId("58e7cf2e46e13dfd851a228a")
created_at: 20141102153003
playback_type: "extended"
file_size_bytes: 0
updated_at: 20141102153003
file_name: "14149710036281.wav"
start_time: 0.692
legacy_video_id: 1228
legacy_user_id: 2423
label: ""
legacy_audio_clip_id: 8906
video: ObjectId("58e7cf2846e13dfd851a18f5")
end_time: 0.692
file_mime_type: "audio/wav"
duration: 19.95
file_path: "/legacy"
user: ObjectId("58e7cf2e46e13dfd851a2289")

```

*Figure 4.5: Database table- audio\_clips*

**4.4.2 audio\_description:** This collection is used to save audio descriptions recorded by describers for the Youtube videos as depicted in Figure 4.6.

```

_id: ObjectId("58e7cf2e46e13dfd851a228a")
status: "published"
legacy_video_id: 1228
legacy_user_id: 2423
audio_clips: Array
 video: ObjectId("58e7cf2846e13dfd851a18f5")
 user: ObjectId("58e7cf2e46e13dfd851a2289")
 language: "en-us"
created_at: 20170317000000
update_at: 20170317000000
notes: ""
likes: 0
overall_rating_average: 4
overall_rating_votes_counter: 7
overall_rating_votes_sum: 31
updated_at: 20190116201350

```

*Figure 4.6: Database table- audio\_description*

**4.4.3 audio\_descriptions\_rating:** This collection is used to save ratings given by blind users to audio description as illustrated in Figure 4.7.

```

_id: ObjectId("591d208b2805421bd3af5918")
audio_description_id: ObjectId("58e7cf3d46e13dfd851a286d")
user_id: ObjectId("58e7e4a8515cd458cef2e017")
created_at: 20170518041819
updated_at: 20170518041819
rating: 3
> feedback: Array
__v: 0

```

*Figure 4.7: Database table- audio\_descriptions\_rating*

4.4.4 **countries**: This collection is used to save country tags in the Youtube videos as shown in Figure 4.8.

```

_id: ObjectId("58bcb31346e13d4cf693f24d")
code: "USA"
id: 1
name: "United States"

```

*Figure 4.8: Database table- countries*

3.4.5 **languages**: This collection is used to save language tags in the Youtube videos as displayed in Figure 4.9.

```

_id: ObjectId("59c584720a3cd1151de62fd6")
code: "af"
name: "Afrikaans"

```

*Figure 4.9: Database table- languages*

4.4.6 **users**: This collection is used to save users in form of their google id as depicted in Figure 4.10.

```

_id: ObjectId("58e7cf2a46e13dfd851a204a")
google_user_id_token: ""
created_at: 20130918144848
legacy_user_id: 25

```

*Figure 4.10: Database table- users*

4.4.7 **users\_votes**: This collection is used to save votes given by users as shown in Figure 4.11.

```
_id: ObjectId("5b87529e5233bd4b6f04916b")
youtube_id: "9Jc7IST4ABI"
user: ObjectId("5b8748f9b85b0d4b690aee55")
updated_at: 20180830021246
created_at: 20180830021246
__v: 0
```

*Figure 4.11: Database table- users\_votes*

4.4.8 **videos**: This collection saves the Youtube video details as reflected in Figure 4.12.

```
_id: ObjectId("58e7cf2846e13dfd851a18f5")
audio_descriptions: Array
description: ""
youtube_id: "B8ISzf2pryI"
views: 0
created_at: 20130617110726
title: "Denver Official Guilty Dog Video www.facebook.com/guiltydog"
legacy_video_id: 1228
updated_at: 20190115232214
```

*Figure 4.12: Database table- videos*

4.4.9 **wish\_list**: This collection saves the video details and votes requested by blind users to be described as illustrated in Figure 4.13.

```
_id: ObjectId("58e41201f7989909dd54a22f")
youtube_id: "GF60Iuh643I"
votes: 2
status: "dequeued"
created_at: 20170404213705
updated_at: 20170816075326
__v: 0
```

*Figure 4.13: Database table- wish\_list*

## **5. Scope and contribution to the project**

The project was done in a team of 7. The website's original version was developed by Rodrigo Leme using ReactJs. The YouDescribe repository is a private Github repository of SKI (Smith-Kettlewell Eye Research Institute) that contains code for the YouDescribe website. A manual was created by me for YouDescribe's existing code base which included the instruction for how to build and run the application. In the beginning of the fall 2019, I helped in User Study 2 in a web application which was to analyze baseline description with AI support and human generated descriptions. Further I learned and used the same technology as used in the development of the YouDescribe version of SKI. Also I started to work on the documentation of the application and learn the API's used. During the semesters, me along with Dr. Ilmi Yoon had several meetings with Prof Pooyan Fazli and Ms. Charity from SKI. The scope of the project escalated from creating a new interface YouDescribeX which accommodates AI support, to creating an admin panel to help admin being independent. Others worked on the backend part and created the API to fetch and store the data which was used on the front end application by me. After documentation and setting goals for the future, I met the professor for weekly updates about the progress of the project and completion of milestones. Each milestone had various challenges and difficulties associated with it but I tried to complete much of my abilities. Like any other project, we are trying to consistently improve on the application with the feedback and usage of the application and the website.

## 6. Implementation

The implementation details are split as YouDescribe website application development using MERN (MongoDB, ExpressJS, ReactJS, NodeJS) stack for both Admin Panel and YouDescribeX, and Caption Rating Game development using HTML, CSS and Javascript.

In the previous semester I implemented the front end for the Admin Panel of the website. This semester I developed front end applications for Caption Rating Game and new Interface YouDescribeX for YouDescribe.

### 6.1 Development Environment

#### 6.1.1 Youdescribe

Build front-end and backend code of Youdescribe website in your local computer:

##### 1) Build front-end

- Modify root\_app/client/src/shared/config.js as follows

```
let apiUrl = `http://127.0.0.1:8080/${apiVersion}`;
```

```
let audioClipsUploadsPath =
```

```
'http://127.0.0.1:8080/audio-descriptions-files';
```

- Modify root\_app/package.json as follows.

```
"scripts": {
```

```
 "start": "NODE_ENV=dev nodemon server/index.js",
```

```
 "build": "webpack --config webpack.config.dev.js"
```

```
},
```

- Open a terminal under root\_app



- Run **npm install** to install the dependencies listed in `root_app/package.json`
- Run **npm run build** to build the project, with a folder generated as `root_app/client/build`
- Run **node server/index.js** (or **npm start**) and the console will be listening to the port 3001

## 2) Build back-end

- Create `root_api/db/config.js` as follows (if not exists)

```
module.exports = {
 hostname: 'localhost',
 database: 'youdescribe',
 username: "",
 password: "",
};
```

- Modify `root_api/index.js` (line 1) as follows

```
const NODE_ENV = "dev";
```

- Open a terminal under `root_api`
- Run **npm install** to install the dependencies listed in `root_api/package.json`
- Run **node index.js** and the console will be listening to the port 8080

## 3) Access Website

- Open a web browser (chrome is recommended)

- Access 127.0.0.1:3001 and should be able to see the YouDescribe website.

### 6.1.2 Caption Rating Game:

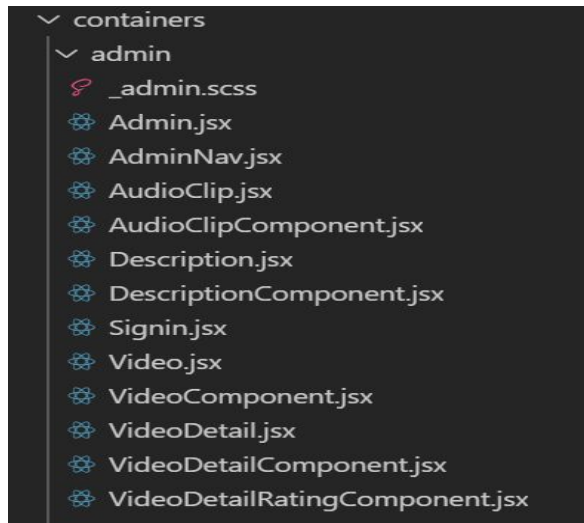
Steps to access the github repository in the local system:

1. Move to the directory in terminal “cd application”
2. Run **npm install** to install the dependencies listed in package.json.
3. Run **node index.js** and the console will be listening to the port 3000.
4. For accessing the website, open any web browser.
5. Access 127.0.0.1:3000 and should be able to see the Caption Rating Game website.

## 6.2 Directory Structure

### 6.2.1 Admin Panel:

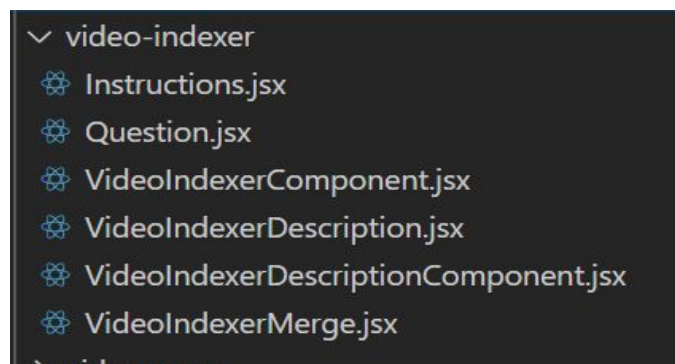
1. Frontend Development:
  - Main focus of the development was to create webpages for the admin panel and make them communicate to the server side.
  - client/ src/ containers/ admin: Contains all the files in which the development of the code as depicted in Figure 6.1.



*Figure 6.1: Directory Structure of Admin Panel*

### 6.2.2 YouDescribeX:

- client/ src/ containers/ video-indexer: Contains all the files in which the development of the code as referred in Figure 6.2 .



*Figure 6.2: Directory Structure of YouDescribeX*

### 6.2.3 Caption Rating Game:

- Webpages were created for the game and made to communicate to the server side. Below are the files in the directory which was used to implement the code as illustrated in Figure 6.3.

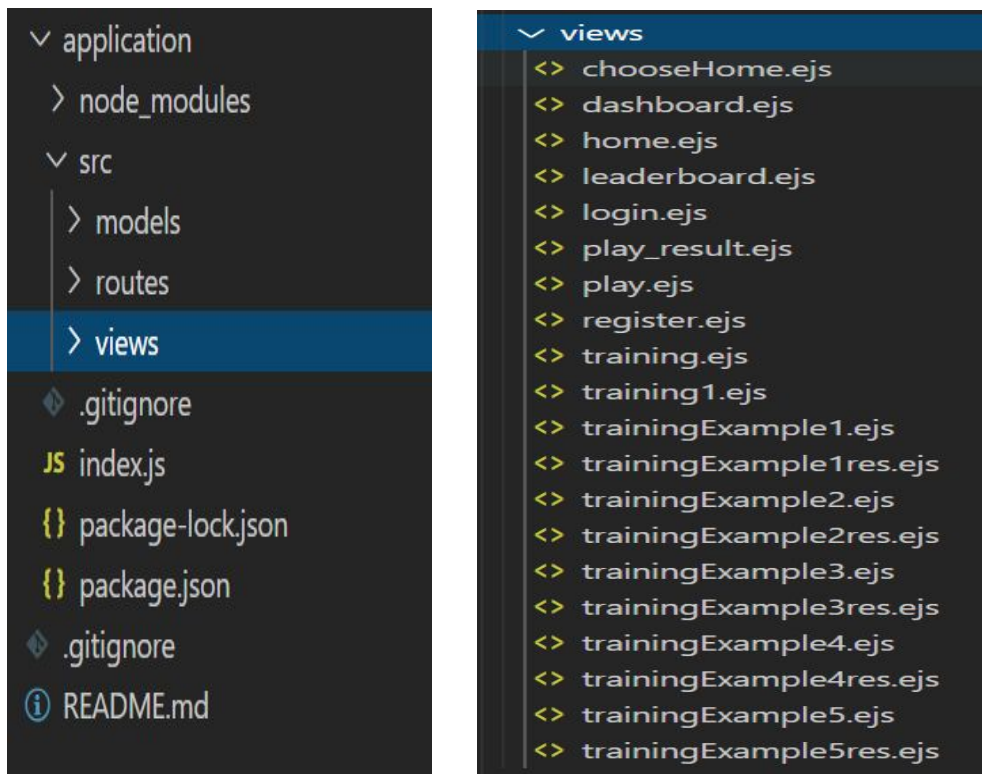


Figure 6.3: Directory Structure of Caption Rating Game

## 6.3 Feature Implementation

### 6.3.1 Admin Panel

1. **Admin Panel Sign In:** Admin will be able to sign in through email id and password that is pre fixed by developers and inform Ms Charity about it.
2. **Admin home page:** Through this admin will be able to see tabs and navigate through different pages in the admin panel.
3. Admin can view the following details on the Descriptions page:
  - **Title:** Title of the Video
  - **Describer:** Name of the describer who described the video.
  - **Description Status:** Status can be published, unpublished and draft.  
This can also be edited by the Admin.

- **Video Status:** Video status can be available/ unavailable depending on the status on youtube.com.
  - **Category:** Category defines the type of video.
  - **Rating:** Rating is overall average rating given by the viewers.
  - **Created At:** It is a timestamp when the video is described to the describer.
4. **Sort:** Admin can sort any of the above fields in ascending as well as descending.
  5. **Search:** Admin can search through Search-by-title, search-by-tag and search-by-describer.
  6. Link redirecting inside the page:
    - **Prev:** Redirecting left hand side on this detail page, to go to the previous video.
    - **Next:** Redirecting right hand side on this detail page, to go to the next video.
    - **Publish:** Admin has access to publish the video to YouDescribe's home page.
    - **Unpublish:** Admin has access to unpublish the video to YouDescribe homepage.
    - **Ratings given by Viewers:** Admin can view all the ratings given by all the viewers.
    - **Feedback:** Click to send an email notification to the describer.

### 6.3.2 YouDescribeX

- 1) **Waitlist Section:** It is a section where the user lands when the video needs time to be preprocessed through Video Indexer. It will also contain some preprocessed video from the wishlist section and sorted based on the most wanted videos demanded by blind viewers that needs to be described. Below is the screenshot Figure 6.4 of the function to fetch the videos from API.

```
fetchingVideosToWaitlist() {
 const url = (`${conf.apiUrl}/wishlist/getbycategories`);
 ourFetch(url).then((response) => {
 const resultArray = response.result;
 resultArray.forEach(resultItem => {
 const category = resultItem._id;

 const youtubeVideosIds = [];
 const dataArray = resultItem.data;
 dataArray.forEach(dataItem => {
 youtubeVideosIds.push(dataItem.youtube_id);
 });
 const ids = youtubeVideosIds.join(',');

 const url = `${conf.youtubeApiUrl}/
 videos?id=${ids}&part=contentDetails,
 snippet,statistics&key=${conf.youtubeApiKey}`;
 ourFetch(url).then(data => {
 this.parseFetchedData(category, data);
 });
 });
 });
 this.closeSpinner();
}
```

Figure 6.4: Function to fetch videos through API.

- 2) **Front-end Implementation:**

**2.a) Scrolling:** This functionality helps to improve the visual interface better and interact because there are many scenes in a single video. To be able to work on the current scene it was necessary to build a scrolling component.

**2.b) Syncing:** The Youtube IFrame API has a function called

`onYouTubeIframeAPIReady()` - This function is automatically called as the page loads and the API has two important callback functions(events):

**2.b.1) `onPlay()`** - The method is called by itself automatically just once when the page loads. This indicates that the player is ready and we can now use the 'player' variable to extract all sorts of information about the video. The player variable can give static information such as 'total duration' of the video and also rapidly changing information such as 'total time elapsed'. From this function, another function is called `updateTime()`.

- **`updateTime()`** - This function is configured to be called after every 100ms i.e., 1/10th of a second. This in turn calls another function if a certain condition is met '`(videotime !== oldTime)`'.
- **`onProgress()`** - This function gives the current time of the video as a parameter. This time value can be used to manipulate the UI. The logic for **highlighting** scenes and auto-pausing video at the end of each scene here.
- **`getSceneFromTime()`** - This function tells which screen is currently being shown to the user on the basis of the current time. This method in turn calls `getIndexOfInputArrForTargetVal()`.
- **`getIndexOfInputArrForTargetVal()`** - This function is an implementation of the binary search algorithm. The function takes 2 inputs:
  - An array of start-times of all scenes
  - Current time of the video

- **scrollToPosition()** - This function contains some boilerplate code for animated/auto scrolling of the list items to a specific position and also some very basic highlighting logic. It requires the position at which the list needs to be scrolled to as an input parameter.

**2.b.1) onStateChange()** - This has been used to update the 'timeAtWhichLastPaused' variable and also to toggle play/pause graphics.

**2.c) Progress bar:** This functionality helps to show a user how far along he/she is in a process to see the scene of the video.

**2.d) Modal:** This functionality is a dialog box/popup window that is displayed on top of the current page and is used to display the results that the user has successfully merged the scenes as shown in Figure 6.5.

```

Modal visible={this.state.visible} width="400" height="200" effect="fadeInUp"
onClickAway={() => this.closeModal()}
 <div>
 <h1 style={{display: "flex", justifyContent: "center", padding: 10}}>Succesfully Merged! </h1>
 <p style={{display: "flex", justifyContent: "center", padding: 10}}>
 You are succesfully able to merge the Scene
 {this.props.scene_num} with {this.props.scene_num + 1}. Thank you!</p>
 <a href="javascript:void(0);" style={{display: "flex", justifyContent: "center", padding: 10}}
 onClick={() => this.closeModal()}>Close
 </div>
</Modal>

```

*Figure 6.5: Modal functionality.*

**2.e) Toast:** It is a feedback message which takes a very little space for displaying while overall activity is interactive and visible to the user. It disappears after a few seconds. This is used to display when the user has saved the descriptions as illustrated in Figure 6.6.



```
toast.success('Saved!', {
 position: "bottom-center",
 autoClose: 1000,
 hideProgressBar: true,
 closeOnClick: true,
 pauseOnHover: true,
 draggable: true,
 progress: undefined,
});
```

Figure 6.6: Toast functionality

### 3) Pass 1:

- It will be easier for the users to determine whether the current scene needs to be merged with the next scene.
- Users will be able to create notes for each scene.
- This will allow the viewers to merge the scene before doing further steps like describing the scene, etc.

### 3.a) Pass 1 API:

#### 3.a.1) Fetch scene data for a video:

- **Request URL:** We need to pass youtube id of the video and user id as the query string parameters:

<http://18.221.192.73:5001/videoData?videoid={youtubeId}&userId={userId}>

- **Scene:** It consists of original scenes produced by Video Indexer for each video.
- **Description:** It consists of scenes of a video modified by a user.
- **Response:** The response json will consist of following as shown in

Figure 6.7:

- **originalData:** This contains original data from table “Scene”

- **modifiedData:** This contains modified data from table “Description”. If the modified data has length > 0 , we use this field otherwise we use original data as shown in Figure 6.8.

**Formatted JSON Data**

```
{
 "youtube_id": "7QZNpS0uos4",
 "indexer_video_id": "4a1a74b4f5",
 "title": "best scene(The Notebook)",
 "duration": 279.446,
 "originalData": [],
 "modifiedData": [],
 "qnaList": []
}
```

*Figure 6.7: Response data from (3.a.1) API*

```
"modifiedData": [
 {
 "sc_id": "7QZNpS0uos4_0_1",
 "sc_num": 0,
 "start_time": 0,
 "end_time": 40.541,
 "original_ocr": "nullnull",
 "original_description": "hello a person abcd wearing a hatRachel McAdams et al. looking",
 "deletedScene": "[]"
 },
 {
 "sc_id": "7QZNpS0uos4_1_4",
 "sc_num": 3,
 "start_time": 40.541,
 "end_time": 156.406,
 "original_ocr": "null",
 "original_description": "a man sitting on a stage\nna man sitting in a dark room\nna perso",
 "deletedScene": "[\n\"7QZNpS0uos4_2_5\"]"
 }
]
```

*Figure 6.8: modifiedData of Response data from (3.a.1) API*

### 3.a.2) Merge of the scenes:

- **URL :** <http://18.221.192.73:5001/saveAiDescription>
- **Method:** POST
- **Body:** {

```
User_id,
videoid,
Scene_arr: [],
qnaData: [scene_id: {question}]
}
```

As shown in Figure 6.9

- Example:** Scene 2 is merged in Scene 1. On the front end, start time, end time, description and ocr is changed for Scene 1.
  - Scene 1      Start Time:  $\text{MIN}(\text{Scene1}, \text{Scene 2})$  start time
  - Scene 1      End time :  $\text{MAX}(\text{Scene 1}, \text{scene 2})$  end time
  - Scene 1      OCR: scene 1 + scene 2
  - Scene 1      Description: scene 1 + scene 2
  - Scene 1      deletedScene: Push scene 2 id in the array.
- The deleted scene array as depicted in Figure 6.10 is used to update a merged\_scene column in the Scene table which has a list of scene Ids, which are merged in a particular scene. As the scene 1 is merged in scene 2, therefore the scene 2 will be deleted from the description table. Description table contains modified data of each user.

```
 user_id: "9fe1e2c6-5cd2-11ea-bc55-0242ac130003"
▶ scene_arr: (28) [{...}, {...}, {...}, {...}, {...}, {...}, {...}, ...
time_taken: 55
▼ qnaData:
▶ 7QZNpS0uos4_0_1: (2) [{...}, {...}]
▶ 7QZNpS0uos4_1_4: [...]
▶ __proto__: Object
videoId: "7QZNpS0uos4"
```

Figure 6.9: Response data from (a.2) API

```
▼ scene_arr: Array(28)
 ▼ 0:
 scene_id: "7QZNp50uos4_0_1"
 modified_description: "a person wearing a hat Rachel McAdams et al. looking"
 modified_ocr: ""
 start_time: 0
 end_time: 6.131
 has_ai: true
 ▶ deletedScene: []
```

Figure 6.10: scene\_arr includes deletedScene from (a.2) API

#### 4) Pass 2 (saving the description):

- After the user has merged all the videos according to the requirements. Users will be able to edit the description part.
- If there is no Text on Screen, the Text on Screen textbox will be hidden.
- If there is text on screen, the Text on Screen textbox will be visible, the user will be able to edit based on the scenes.

#### 4.a) Pass 2 API:

##### 4.a.1) Save modified scene data into the Database

- **URL :** <http://18.221.192.73:5001/saveAiDescription>
- **Method:** POST
- **Body:** {  
    User\_id,  
    videoid,  
    Scene\_arr: [ ],  
    qnaData: [ scene\_id: {question} ]  
}

As displayed in Figure 6.11

- The api iterates over the scene array and inserts scene data if not present, and if present, it updates the scene. This can be used to update one or more than one scene.

- **Response**

- On success: status 200, success message
- On failure: error will be printed



```

▼ scene_arr: Array(28)
 ▼ 0:
 scene_id: "7QZNpS0uos4_0_1"
 modified_description: "a person wearing a hat"
 modified_ocr: "Rachel McAdams et al. lookin"
 start_time: 0
 end_time: 6.131
 has_ai: true
 deletedScene: []

```

*Figure 6.11: Save modified scene data from (4.a.1) API*

## 5) Pass 2 (converting text to speech):

- The describer is given a chance to see what the actual described video looks like.
- Users won't be able to edit the dialog timeline of the video but will be able to drag the description timelines.
- Users will have a pre suggestion of inline or extended description. Further they can also change description timeline depending on the requirement.
- It will be displayed scene by scene and the user will work on one scene at a time.

### 5.a) Pass 2 API:

#### 5.a.1) Get mp3 audio file of a text/description

- **Url:** <http://18.221.192.73:5001/descriptionMp3Audio>

- **Method:** Post
- **Body:** {
 

```

 inputText: 'Hello. This is to check the mp3 output for generating tts'

```
- **Response:** Mp3 audio file.

#### 5.a.2) Get All information regarding audio descriptions for a video:

This API will run the algorithm where the descriptions are divided into sentences and the audio clip duration is used to determine whether the audio description should be inline/extended.

- **Url:** <http://18.221.192.73:5001/generateAudioDescriptionMp3>
- **Method:** GET
- **Parameters:** videoId, volunteerId
- **API Call Example:**

```

http://18.221.192.73:5001/generateAudioDescriptionMp3?videoId=<video id>&volunteerId=<user id>

http://18.221.192.73:5001/generateAudioDescriptionMp3?videoId=7QZNpS0uos4&volunteerId=9fe1e2c6-5cd2-11ea-bc55-0242ac130003

```
- **Response Parameters:** Response is a JSON body which has the following parameters for every scene
  - Description - Description for the scene
  - Sentences - An array of sentences for the scene description
  - Empty gaps - Empty gaps timestamps where audio description can be played inline.
- Each sentence in the sentences array has the following parameters as shown

in Figure 6.12,

- "sentence id": unique sentence id
- "sentence": sentence in text
- "audio type": inline/extended
- "audio start time": audio's starting time stamp
- "audio duration": audio clip's duration
- "Url": The url for the location of the '.mp3' file
- **Response Example:**

```
{
 "7QZNpS0uos4_0_1": {
 "description": "hello I a person abcd wearing a hatRachel McAdams et al. looking at the cameraa close",
 "sentences": [
 {
 "sentence id": "9fe1e2c6-5cd2-11ea-bc55-0242ac130003_7QZNpS0uos4_ai_0_0",
 "sentence": "hello I a person abcd wearing a hatRachel McAdams et al.",
 "audio type": null,
 "audio start time": null,
 "audio duration": 5.064,
 "url": "http://18.221.192.73:5001/audio/7QZNpS0uos4/ai/9fe1e2c6-5cd2-11ea-bc55-0242ac130003/0/0"
 },
 { },
 { }
],
 "empty gaps": [
]
 },
 "7QZNpS0uos4_1_4": {
 "description": "a man sitting on a stage\na man sitting in a dark room\na person wearing a costume\na",
 "sentences": [
 {
 "sentence id": "9fe1e2c6-5cd2-11ea-bc55-0242ac130003_7QZNpS0uos4_ai_3_0"
```

*Figure 6.12: Response data from (5.a.2) API*

## 7. Results

### 7.1 Admin Panel

1) **Admin Panel Sign In:** As depicted in Figure 7.1

**URL:** <http://dev.youdescribe.org:3001/admin/signin>

The screenshot shows a web browser window with the address bar displaying "Not secure | 52.53.151.37:3000/admin/signin". The page header includes the "YD YOUDESCRIBE" logo, a search bar, and links for "RECENT DESCRIPTIONS", "WISH LIST", "SUPPORT", and "Sign in with Google". The main content area contains a login form with the following elements:

- Email address:** A text input field with the placeholder "Enter email".
- Password:** A text input field with the placeholder "Enter password".
- Submit:** A blue button labeled "Submit".

*Figure 7.1: Sign in of Admin Panel page*

2) **Admin home page:** As shown in Figure 7.2

**URL:** <http://dev.youdescribe.org:3001/admin>

The screenshot shows the Admin Panel Home page. The top navigation bar is blue and contains the text "ADMIN PANEL". Below the navigation bar, there are three main sections: "Videos", "Descriptions", and "Statistics". Each section has a search bar and a "Search" button. The "Videos" section is currently active. Below the navigation bar, there is a "List of Videos" table. The table has the following columns: "Id", "Title", "Category", "Status", "Created At", and "Action". The table displays five video entries.

| Id       | Title                                                   | Category              | Status    | Created At | Action                          |
|----------|---------------------------------------------------------|-----------------------|-----------|------------|---------------------------------|
| c3 added | The Simpsons: The Simpsons in a indoor Waterpark        | People & Blogs        | available | 2019-09-29 | <a href="#">Video Remapping</a> |
| a292dd   | How to Create the Perfect Cheese Platter    KIN EATS    | Howto & Style         | available | 2019-09-27 | <a href="#">Video Remapping</a> |
| c3 added | Illinois State University   Sober Redbirds              | People & Blogs        | available | 2019-09-27 | <a href="#">Video Remapping</a> |
| c3 added | Claire   The Documentary                                | Nonprofits & Activism | available | 2019-09-27 | <a href="#">Video Remapping</a> |
| a292cc   | Samsung India Good Vibes App: Caring for the impossible | Science &             | available | 2019-09-26 | <a href="#">Video Remapping</a> |

*Figure 7.2: Home page of Admin Panel*



- 3) **Description:** Admin can view the following details on the Descriptions page as referred in Figure 7.3:

**URL:** <http://dev.youdescribe.org:3001/admin/descriptions>

ADMIN PANEL

Videos

Find Videos...

Search

Descriptions

Find Descriptions...

Search

Statistics ▾

SignOut

Click Id for Full Info

Click Column Header for Sorting

1 - 50 out of 3154

| Id                     | Title                                                | Describer     | Description Status | Video Status | Video Category | Overall Rating | Created At |
|------------------------|------------------------------------------------------|---------------|--------------------|--------------|----------------|----------------|------------|
| <a href="#">c3fdd6</a> | The Simpsons: The Simpsons in a indoor Waterpark     | Momo Heart    | draft              | available    | People & Blogs | 0              | 2019-09-29 |
| <a href="#">a292dc</a> | How to Create the Perfect Cheese Platter    KIN EATS | Chad Hinton   | draft              | available    | Howto & Style  | 0              | 2019-09-27 |
| <a href="#">c3fdc2</a> | Illinois State University   Sober Redbirds           | Hunter Thomas | published          | available    | People & Blogs | 4              | 2019-09-27 |
| <a href="#">c3fdbe</a> | Claire   The Documentary                             | Rev Samuelson | draft              | available    | Nonprofits &   | 0              | 2019-09-27 |

*Figure 7.3: Description page of Admin Panel*

- 4) **Sort:** Admin can sort any of the fields in ascending as well as descending:

**Example:**Sorting based on Description Status as displayed in Figure 7.4.

| List of Descriptions   |                                             |                                   |                    |              |                  |        |            |
|------------------------|---------------------------------------------|-----------------------------------|--------------------|--------------|------------------|--------|------------|
| Id                     | Title                                       | Describer                         | Description Status | Video Status | Category         | Rating | Created At |
| <a href="#">af593a</a> | Key & Peele - Retired Military Specialist   | Rupal Khilari                     | draft              | available    | Comedy           |        | 2017-05-18 |
| <a href="#">af594b</a> | Ed Sheeran - Galway Girl [Official Video]   | Logan and Samuel the Epic channel | draft              | available    | Music            |        | 2017-05-18 |
| <a href="#">af5978</a> | Pawnee Indian fight's with an Us cavalryman | Marik                             | draft              | unavailable  |                  |        | 2017-05-18 |
| <a href="#">af59e2</a> | Too Quick To Judge (Touching Short-Film)    | Rupal Khilari                     | draft              | available    | Film & Animation |        | 2017-05-27 |

*Figure 7.4: Sorting based on Description status*

**Example:** Sorting based on Ratings as illustrated in Figure in 7.5

### List of Descriptions

| Id     | Title                                                          | Describer         | Description Status | Video Status | Category       | Rating | Created At |
|--------|----------------------------------------------------------------|-------------------|--------------------|--------------|----------------|--------|------------|
| 42f739 | Degrassi Junior High (S. 1 Ep. 1) – Teen Drama                 | Melissa Hope      | published          | available    | Entertainment  | 5      | 2019-03-27 |
| 54d0f9 | The uncomplicated truth about women's sexuality   Sarah Barmak | Sean Shaifubahrim | published          | available    | People & Blogs | 5      | 2019-03-22 |
| 19dfe0 | BI 101 – "Unit 3, Lecture 1 (Evolution)" (Periard)             | GRCctv            | published          | available    | Education      | 5      | 2019-03-15 |

Figure 7.5: Sorting based on Ratings

### 5) Search:

- a) Admin can search any description through Search-by-title, search-by-describer, search-by-description-status, search-by-video-status and search-by-video-category .

**Example:** Searching based on Describers name as illustrated in Figure 7.6.

**URL:** <http://dev.youdescribe.org:3001/admin/descriptions?keyword=enoch&order=&sortby=>

|        |                                             |                                       |              |                                    |                                       |              |                         |
|--------|---------------------------------------------|---------------------------------------|--------------|------------------------------------|---------------------------------------|--------------|-------------------------|
| Videos | <input type="text" value="Find Videos..."/> | <input type="button" value="Search"/> | Descriptions | <input type="text" value="enoch"/> | <input type="button" value="Search"/> | Statistics ▾ | <a href="#">SignOut</a> |
|--------|---------------------------------------------|---------------------------------------|--------------|------------------------------------|---------------------------------------|--------------|-------------------------|

| List of Descriptions            |                                                                  |           |                    |              |                       |                |            |
|---------------------------------|------------------------------------------------------------------|-----------|--------------------|--------------|-----------------------|----------------|------------|
| Click Id for Full Info          |                                                                  |           |                    |              |                       |                |            |
| Click Column Header for Sorting |                                                                  |           |                    |              |                       |                |            |
|                                 |                                                                  |           |                    |              |                       |                |            |
| Id                              | Title                                                            | Describer | Description Status | Video Status | Video Category        | Overall Rating | Created At |
| <a href="#">7c633e</a>          | The Snip - Simon's Cat   SHORTS #81                              | Enoch Lee | draft              | available    | Film & Animation      | 0              | 2019-06-05 |
| <a href="#">7c633a</a>          | The Snip - Simon's Cat   SHORTS #81                              | Enoch Lee | published          | available    | Film & Animation      | 5              | 2019-06-05 |
| <a href="#">f93e84</a>          | Baby Gorilla Born at Disney's Animal Kingdom   Walt Disney World | Enoch Lee | published          | available    | Travel & Events       | 5              | 2019-06-03 |
| <a href="#">330ea5</a>          | Pip   A Short Animated Film                                      | Enoch Lee | published          | available    | Nonprofits & Activism | 5              | 2019-06-10 |
| <a href="#">916063</a>          | FLUNNY and CUTE SOULBLES will make you                           | Enoch Lee | published          | available    | Pets & Animals        | 4              | 2019-04-06 |

Figure 7.6: Searching based on Descriptor name

- b) Admin can search any video through search-by-title, search-by-id, search-by-status and search-by-category.

**Example:** Searching based on Videos keywords as illustrated in Figure 7.7.

**URL:** [http://dev.youdescribe.org:3001/admin/videos?keyword=science&order=-1&sortby=\\_id](http://dev.youdescribe.org:3001/admin/videos?keyword=science&order=-1&sortby=_id)

The screenshot shows the 'Videos' admin interface. At the top, there's a search bar with 'science' entered and a 'Search' button. Below this is a table titled 'List of Videos' with 6 columns: Id, Title, Category, Status, Created At, and Action. The table contains 5 rows of video data. The first row has the ID 'a292cc' and title 'Samsung India Good Vibes App: Caring for the impossible', categorized under 'Science & Technology'. The other rows show videos related to 'Fire Rescue', 'Inclusion Starts With I', '2 easy science experiments', and 'UWF Summer Festival Chorus'.

| Id                     | Title                                                   | Category             | Status    | Created At | Action                          |
|------------------------|---------------------------------------------------------|----------------------|-----------|------------|---------------------------------|
| <a href="#">a292cc</a> | Samsung India Good Vibes App: Caring for the impossible | Science & Technology | available | 2019-09-26 | <a href="#">Video Remapping</a> |
| <a href="#">be5022</a> | Fire Rescue Science Technology                          | Education            | available | 2019-09-13 | <a href="#">Video Remapping</a> |
| <a href="#">be4ecf</a> | Inclusion Starts With I (Voiceover)                     | Science & Technology | available | 2019-08-20 | <a href="#">Video Remapping</a> |
| <a href="#">36069b</a> | 2 easy science experiments [pt 3]                       | Howto & Style        | available | 2019-08-01 | <a href="#">Video Remapping</a> |
| <a href="#">7ebec4</a> | UWF Summer Festival Chorus: Choral Masterpieces         | Education            | available | 2019-07-25 | <a href="#">Video Remapping</a> |

*Figure 7.7: Searching based on Videos keywords*

- 6) **Link redirecting inside the description page:** As seen in Figure 7.8

**URL:** [http://dev.youdescribe.org:3001/admin/description/detail/5d90131ad5be181566c3fdd6?keyword=&order=-1&sortby=video.\\_id](http://dev.youdescribe.org:3001/admin/description/detail/5d90131ad5be181566c3fdd6?keyword=&order=-1&sortby=video._id)

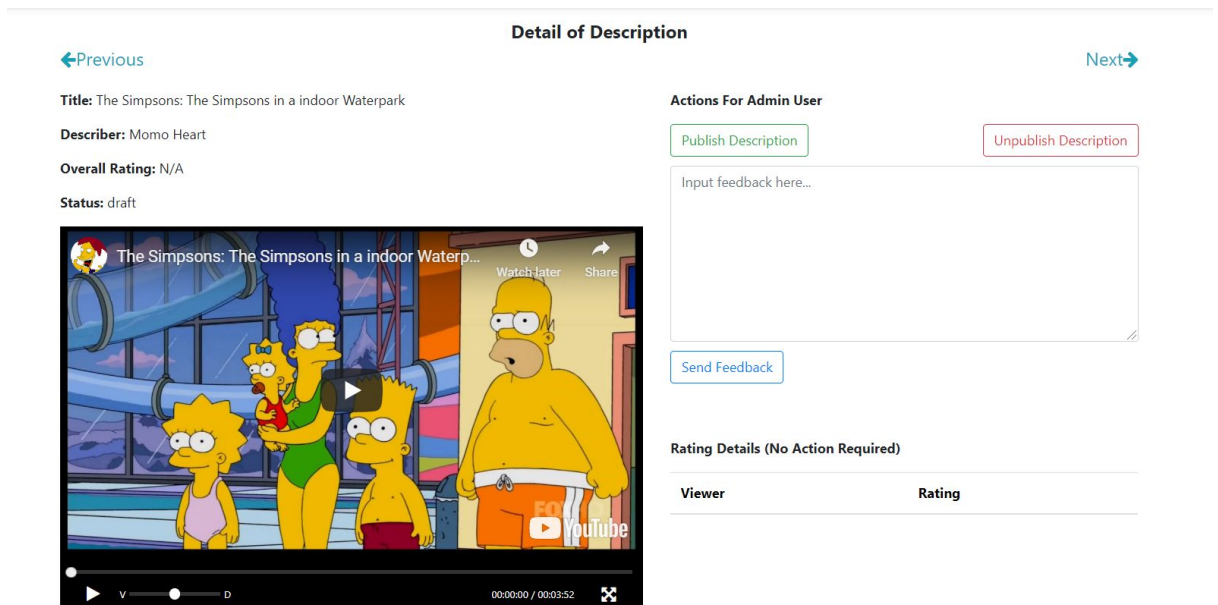


Figure 7.8: Detailed description page of Admin Panel

7) **Video Page for admin access:** As reflected in Figure 7.9.

**URL:** <http://dev.youdescribe.org:3001/admin/video>

| ADMIN PANEL                     |                                                         |                       |              |                      |                 |                      |
|---------------------------------|---------------------------------------------------------|-----------------------|--------------|----------------------|-----------------|----------------------|
| Videos                          | Find Videos...                                          | Search                | Descriptions | Find Descriptions... | Search          | Statistics ▾ SignOut |
| List of Videos                  |                                                         |                       |              |                      |                 |                      |
| Click Id for Full Info          |                                                         |                       |              |                      |                 |                      |
| Click Column Header for Sorting |                                                         |                       |              |                      |                 |                      |
| 1 - 50 out of 2709              |                                                         |                       |              |                      |                 |                      |
| Id                              | Title                                                   | Category              | Status       | Created At           | Action          |                      |
| c3fdd7                          | The Simpsons: The Simpsons in a indoor Waterpark        | People & Blogs        | available    | 2019-09-29           | Video Remapping |                      |
| a292dd                          | How to Create the Perfect Cheese Platter    KIN EATS    | Howto & Style         | available    | 2019-09-27           | Video Remapping |                      |
| c3fdc3                          | Illinois State University   Sober Redbirds              | People & Blogs        | available    | 2019-09-27           | Video Remapping |                      |
| c3fdbf                          | Claire   The Documentary                                | Nonprofits & Activism | available    | 2019-09-27           | Video Remapping |                      |
| a292cc                          | Samsung India Good Vibes App: Caring for the impossible | Science &             | available    | 2019-09-26           | Video Remapping |                      |

Figure 7.9: Video page of Admin Panel

8) **Statistics page:** As shown in Figure 7.10

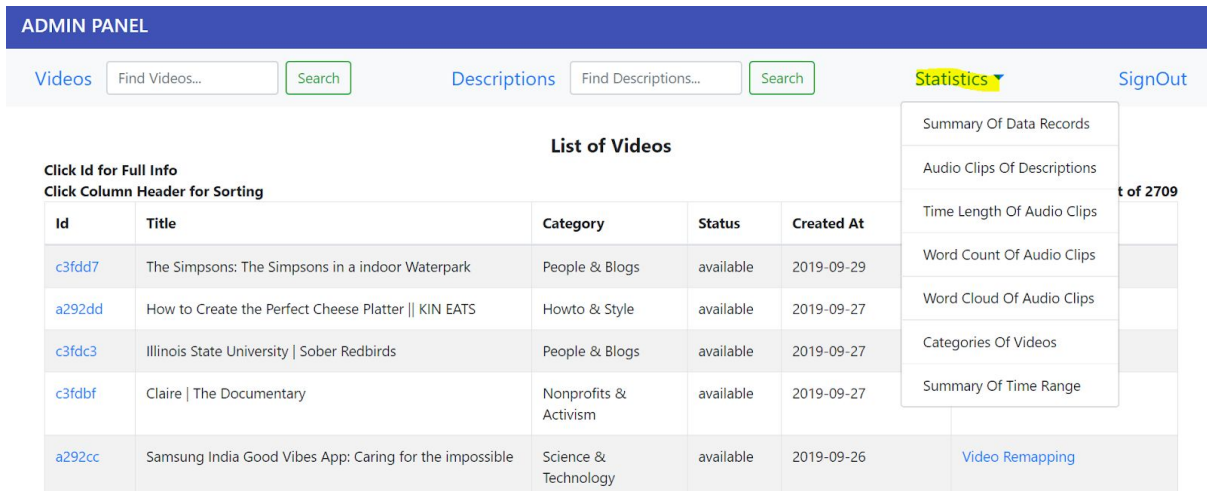


Figure 7.10: Statistics page of Admin Panel

- **Data Records:** Shows number of records which are used to analyze statistics as illustrated in Figure 7.11.

**URL:** <http://dev.youdescribe.org:3001/statistics/summaryofdatarecords>

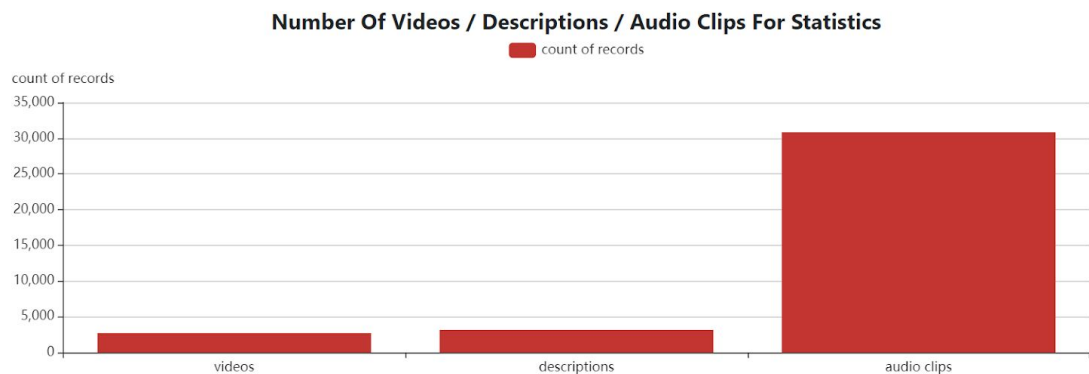
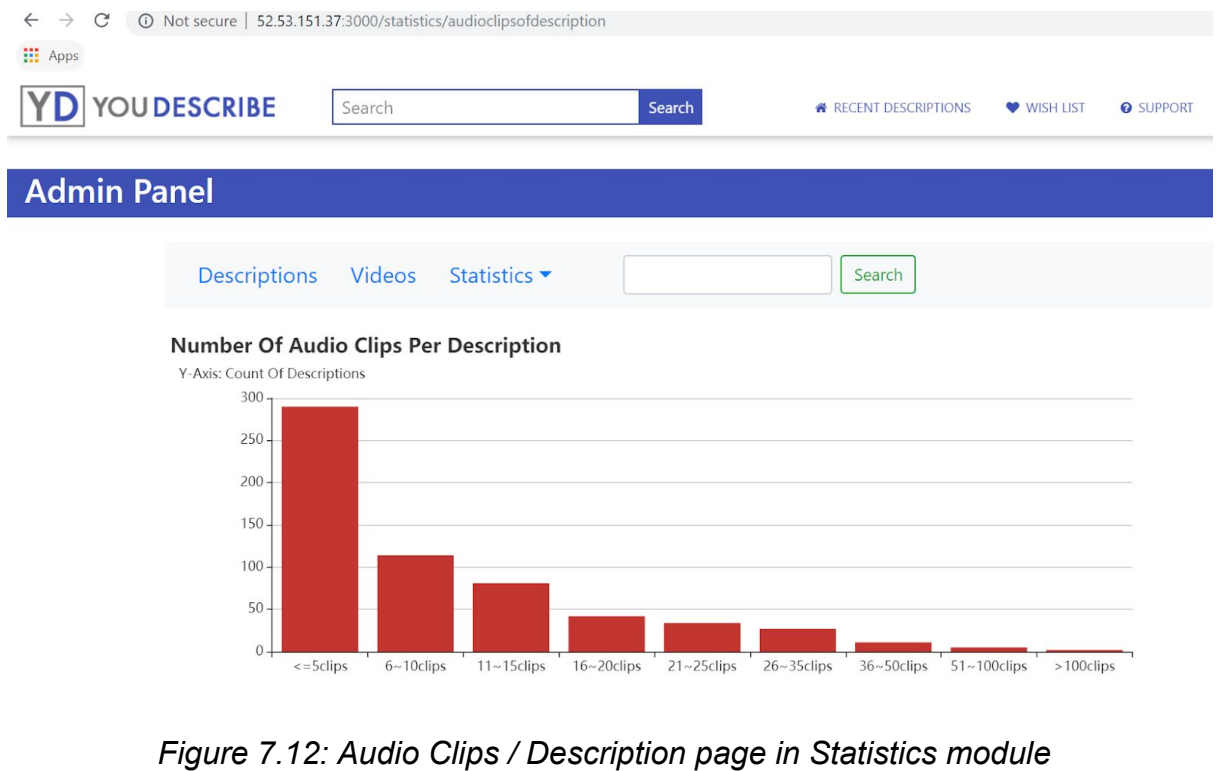


Figure 7.11: Data records page in Statistics module

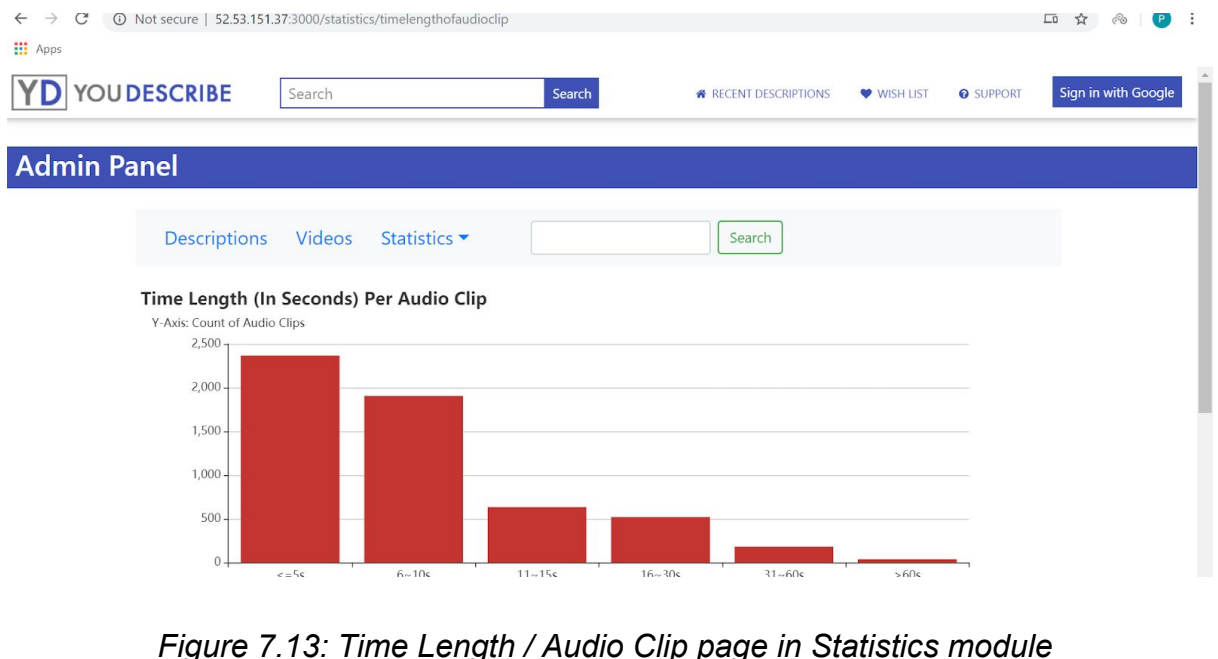
- **Audio Clips / Description:** As seen in Figure 7.12

**URL:** <http://dev.youdescribe.org:3001/statistics/audioclipsofdescription>



- **Time Length / Audio Clip:** As depicted in Figure 7.13

**URL:**<http://dev.youdescribe.org:3001/statistics/timelengthofaudioclip>



- **Word Count / Audio Clip:** As seen in Figure 7.14

**URL:**<http://dev.youdescribe.org:3001/statistics/wordcountofaudioclip>

### Number Of Words Per Audio Clip

Y-Axis: Count of Audio Clips

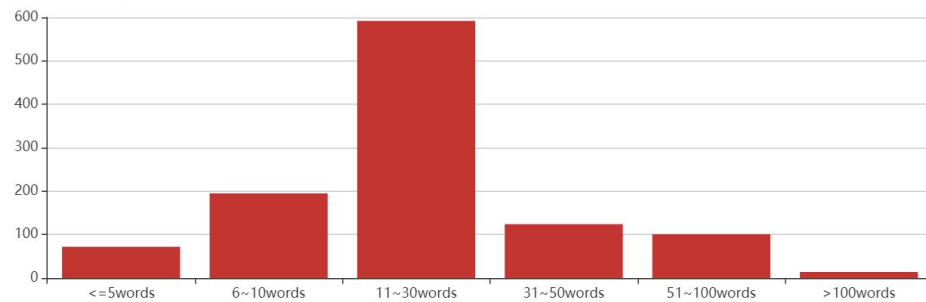


Figure 7.14: Word Count / Audio Clip page in Statistics module

## 7.2 YouDescribeX

1. **To access YouDescribeX:** As demonstrated in Figure 7.15

**URL:** <http://dev.youdescribe.org:3001/authoring-tool/dJJ0yadpqKI>

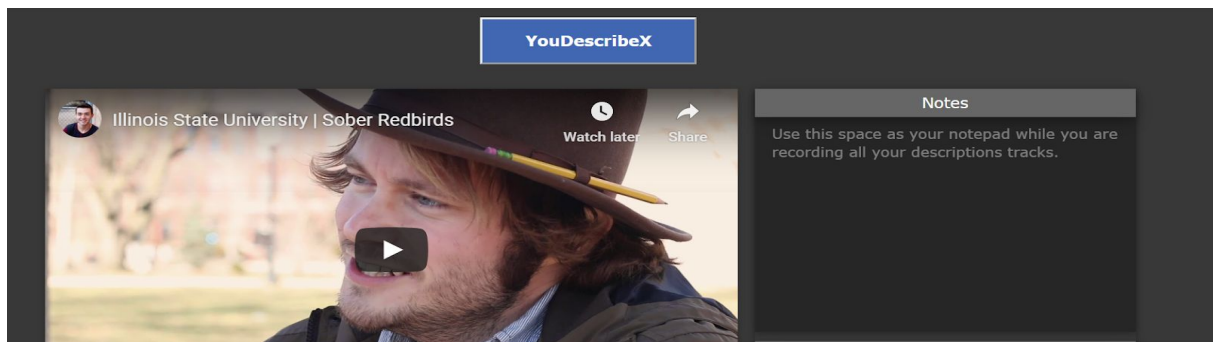


Figure 7.15: Access YouDescribeX

2. **Waitlist:** If the video is not preprocessed by VideoIndexer API, it will go to the waitlist section. Suggestions are coming based on categories of most demanding videos see Figure 7.16.

**URL:** <http://dev.youdescribe.org:3001/waitlist>



Thank You for choosing the new version of YOUDESCRIBE.

Thank you for checking out YouDescribeX. Because this is an AI assisted platform the algorithm needs some time to process your video before \*you\* can get to work! Right now, because this technology is still being trained, it takes around 30 minutes to process a video. YouDescribeX will send you an email when it is ready to describe. Leaving you free to work on something else.

If you had some time, the videos below have been requested from our viewers and are processed, ready for a human to expand, correct, and simplify. Less than xx % of wish list items get described, so if you have a few minutes while your video processes, I am sure a viewer would be thrilled if you could describe their request!

Suggestions for you based on Categories:

Education



6:25

No Mirrors in My Nana's House read b...

Author: StorylineOnline



7:45

Deaf Discrimination on ABC's 'What Would...'

Author: SC



2:43

My mum's phone

Author: Paul Andres



5:01

Prodigi Connect 12 from Humanware

Author: PATINSProject

Suggestions for you based on Categories:

Education



6:25

No Mirrors in My Nana's House read b...

Author: StorylineOnline



7:45

Deaf Discrimination on ABC's 'What Would...'

Author: SC



2:43

My mum's phone

Author: Paul Andres



5:01

Prodigi Connect 12 from Humanware

Author: PATINSProject

Entertainment



8:43

Goldfische: Warum sind sie beliebt und wie...

Author: Galileo



13:12

Coldplay's FULL Pepsi Super Bowl 50 Halfti...

Author: NFL



1:38

Pittsburgh International Festival ...

Author: PGH Cultural Trust



1:31

Check Out CRITICAL ROLE's New Intro!

Author: Geek & Sundry

People & Blogs





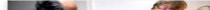




Figure 7.16: Waitlist section of YouDescribeX

3. Instructions: As illustrated in Figure 7.17



## Instructions:

Your YouTube video has been processed! A series of computer driven algorithms have made some initial descriptions and broken down the video into segments.

**Step 1** Please watch the video through once, giving special attention to the scene pacing. Your first job is to merge any scenes that are too choppy for good audio description. There is a note taking section for any random thoughts or words you might want to use on your next viewing.

**Step 2** After merging scenes as best you can, you can start in on writing the script into the text box. Some computer generated sample text is there to help guide you. Speech to text capability means you won't need a microphone to record your own voice, but it does mean you need to be careful about typos and punctuation!



Figure 7.17: Instructions to use YouDescribeX

## 4. Pass 1: Merging the scenes

URL: <http://dev.youdescribe.org:3001/video-indexer-merge>

Each scene is highlighted so that the user will be able to understand which scene is the current scene to work on as seen in Figure 7.18.

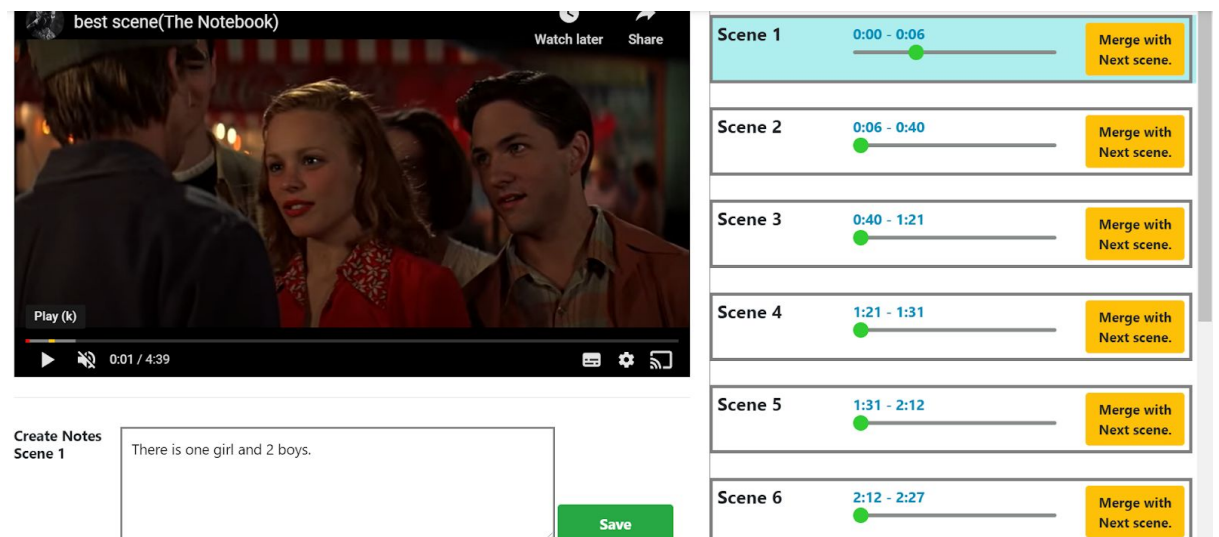


Figure 7.18: Highlight scene synced with video and notes in  
YouDescribeX

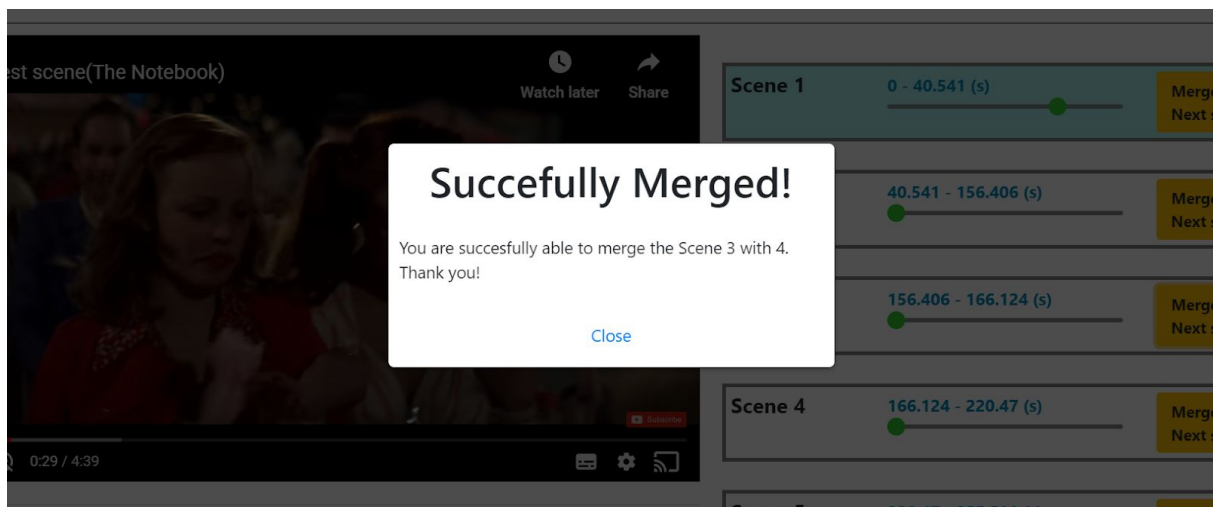
## 5. Clicking on the merge button will merge the scenes.

**Example** : Here scene 3 is merged with scene 4 as shown in Figure 7.19.

Results displayed using Modal as reflected in Figure 7.20.



*Figure 7.19: Merging of the scene in YouDescribeX*



*Figure 7.20:Modal popup in YouDescribeX*

- Pass 2: Saving the description and text on screen into the database:** as reflected in Figure 7.21

**URL:** <http://dev.youdescribe.org:3001/video-indexer-description>

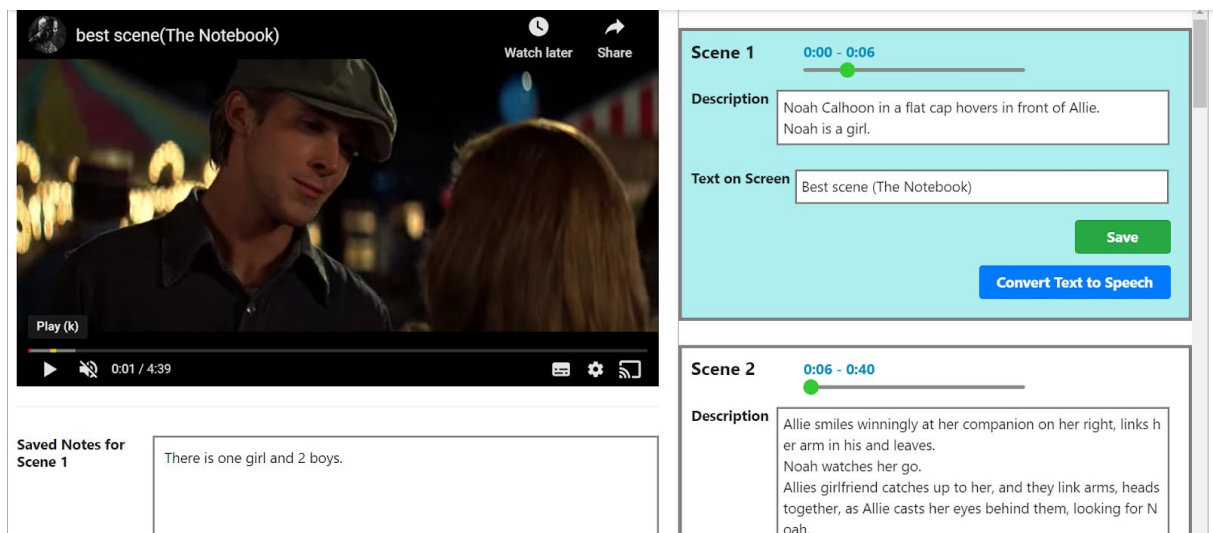


Figure 7.21: Saving the descriptions in YouDescribeX

7. Clicking on the **Save** button will save the data into the database. Toast displayed to confirm the saving as depicted in Figure 7.22.



Figure 7.22: Toast confirmation of saving the description

8. Clicking on the **Convert Text to Speech** button will convert the descriptions into timelines and highlight the selected sentence as shown in figure 7.23.

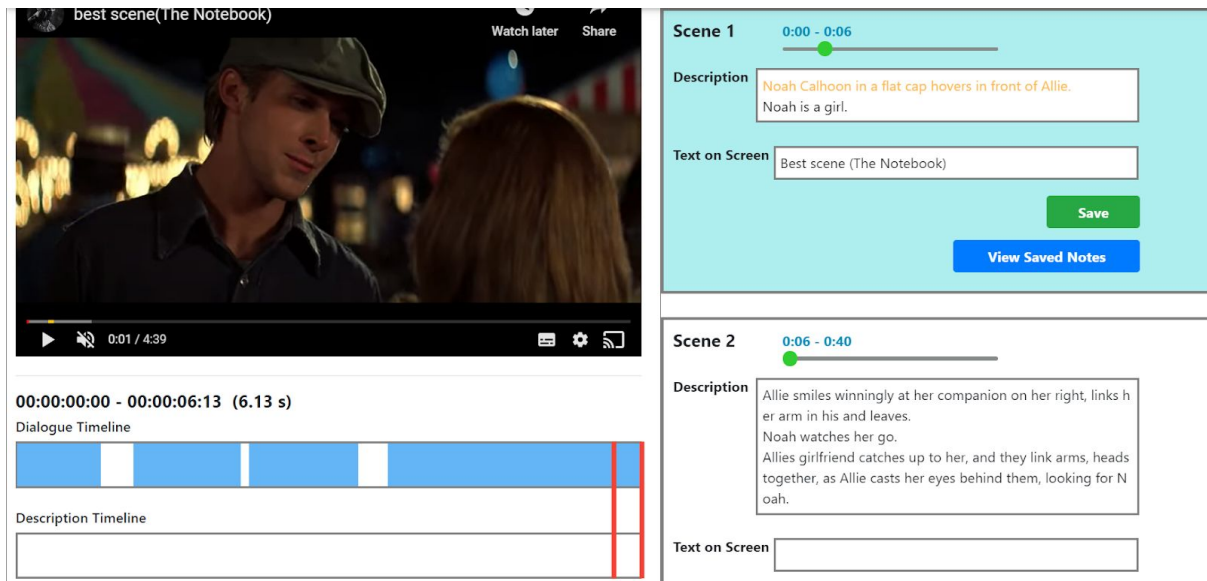


Figure 7.23: Convert Text to Speech and highlight the selected sentence.

9. Users will be able to choose between inline or extended descriptions as shown in Figure 2.24.

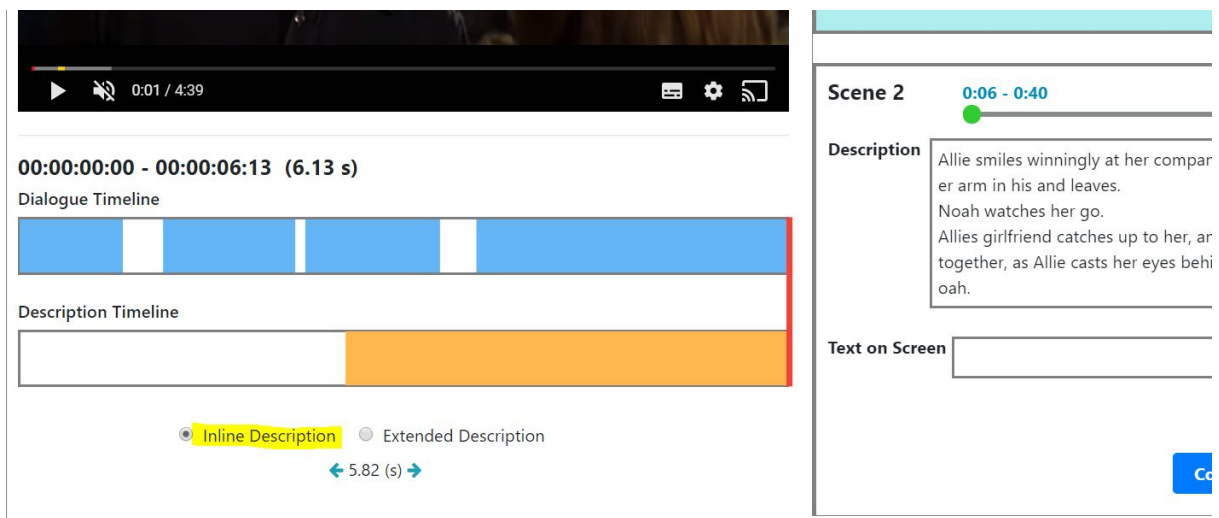
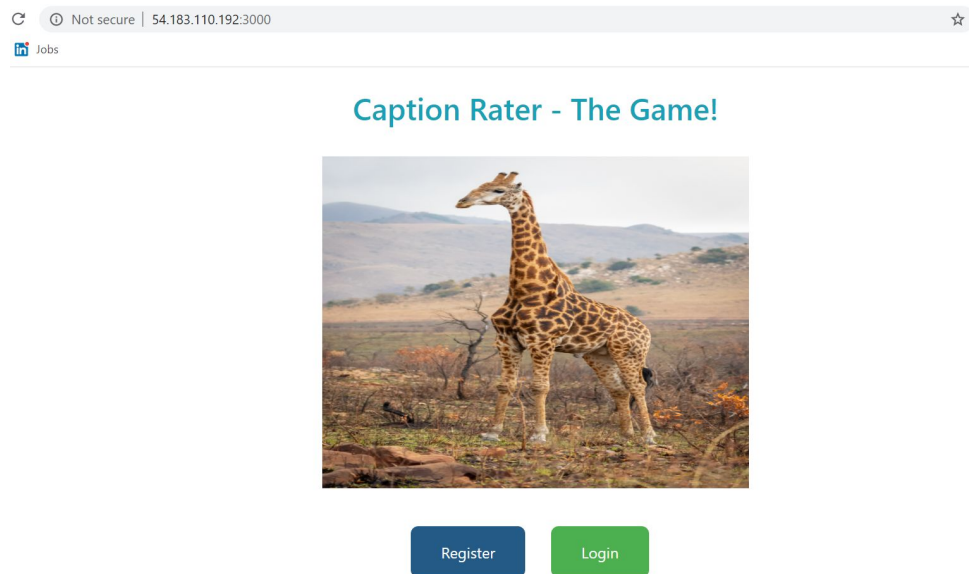


Figure 7.24: Change selected description to inline or extended descriptions.

## 7.3 Caption Rating Game

1. **Home Page:** As depicted in Figure 7.25

**URL:** <http://54.183.110.192:3000/>



*Figure 7.25: Home Page of Caption Rating Game*

2. **Register page:** As seen in Figure 7.26

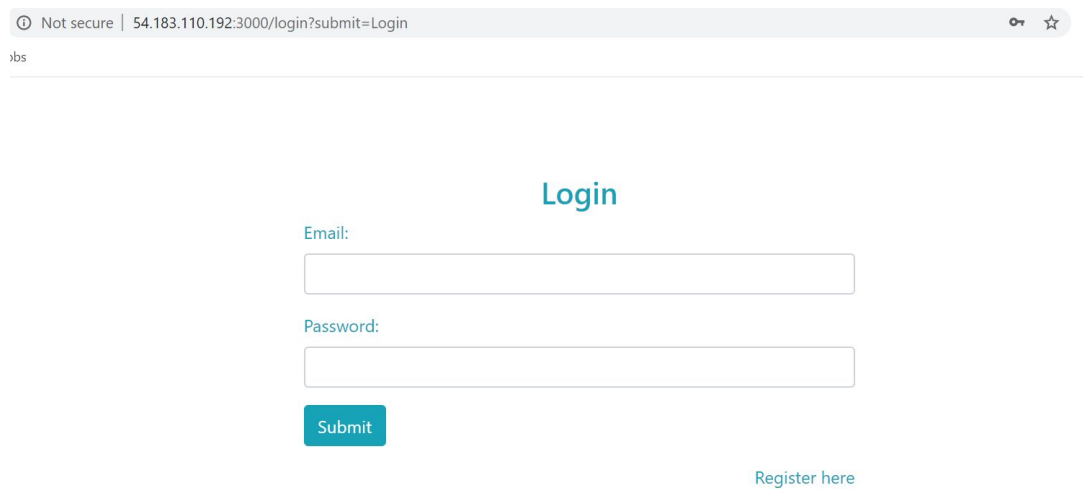
**URL:** <http://54.183.110.192:3000/register?submit=Register>

A screenshot of a web browser showing the register page. The browser's address bar shows "Not secure | 54.183.110.192:3000/register?submit=Register". The main content area features the title "Register" in blue text. Below the title, there are four input fields with labels: "Username:", "Email:", "Password:", and "Confirm Password:". Each label is in blue text and is positioned to the left of its corresponding input field. At the bottom of the form, there is a blue "Submit" button.

*Figure 7.26: Register Page of Caption Rating Game*

3. **Login:** As illustrated in Figure 7.27

**URL:** <http://54.183.110.192:3000/login?submit=Login>



Not secure | 54.183.110.192:3000/login?submit=Login

## Login

Email:

Password:

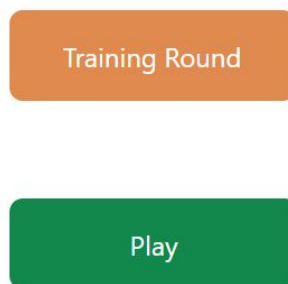
Submit

[Register here](#)

*Figure 7.27: Login Page of Caption Rating Game*

4. **Choose home page:** As shown in Figure 7.28

## Caption Rater - The Game!



Training Round

Play

*Figure 7.28: Choose Home Page of Caption Rating Game*

5. **Training Rounds:** As reflected in Figure 7.29

**URL:** <http://54.183.110.192:3000/training?submit=Training+Round>

## Caption Rater - The Game!

You will be presented with a series of images, along with a caption that describes the image.

Your goal: Rate how well the caption fits the image, using the scale below. The closer your rating matches to a group consensus, the more points you earn!

| Rating | Description                                       |
|--------|---------------------------------------------------|
| 1      | Poor caption. Completely wrong.                   |
| 2      | Captures part of image                            |
| 3      | Captures part of image, weird language or action. |
| 4      | Describes image mostly.                           |
| 5      | Describes the image perfectly!                    |

Next

*Figure 7.29: Instructions in Training Round of Caption Rating Game*

### 6. Training rounds for the game: As depicted in Figure 7.30

## Caption Rater - The Game!

Instructions



A woman wearing a net on her head cutting a cake.

1 2 3 4 5

Submit

*Figure 7.30: Training of Caption Rating Game*



## 7. Results of the training round: As seen in Figure 7.31



Whomp  
whomp!

Your rating: 3  
Consensus: 4  
Scores Earned: 5

A woman wearing a net on her head cutting a cake.

Explanation for the consensus: We expect to receive more explanation about the background of the image!

Next

Figure 7.31: Results of Training Round in Caption Rating Game

## 8. Play round: As observed in Figure 7.32

Not secure | 54.183.110.192:3000/play\_result?comment=So%20close!%20Try%20harder%20next%20time!&rate=3&score=10&consensus=2...

ibs

A photograph of a white bus with blue and green accents, parked on a city street. The bus is facing towards the left of the frame. In the background, there are buildings, trees, and a street sign.

A bus that is sitting in the street.

So close! Try  
harder next  
time!

Your rating: 3  
Consensus: 2.8750  
Scores Earned:10

Next

dashboard

Figure 7.32: Play section of Caption Rating Game



## 9. Dashboard: As demonstrated in Figure 7.33

URL: <http://54.183.110.192:3000/dashboard?submit=dashboard>

Not secure | 54.183.110.192:3000/dashboard?submit=dashboard

bs

### Caption Rater - The Game!

#### Dashboard

Total Score: 10

Accuracy: 50.00 %

Number of Attempts: 1

| Image                                                                             | Caption                              | Your Rating | Consensus | Scores earned | Create a dispute   |
|-----------------------------------------------------------------------------------|--------------------------------------|-------------|-----------|---------------|--------------------|
|  | A bus that is sitting in the street. | 3           | 2.8750    | 10            | <div>Dispute</div> |

Play

*Figure 7.33: Dashboard of Caption Rating Game*

## **8. Conclusion and Future Work**

### **8.1 Conclusion**

The objective of the YouDescribe is to help admin to be able to work independently, analyze the data for improvement, help describers to use AI generated descriptions and generate automatic speech descriptions. Caption Rating Game is to collect data for captions and images to train a dataset to give more accurate results.

The goal of this MS project is to upgrade the website YouDescribe. The importance of this document is to record the work done in this project so that future students will be able to quickly understand the work and advance it further. The innovative functionalities offered by the app such as text to speech and auto generated descriptions make descriptions less time consuming and more enjoyable. With features like inline descriptions and baseline descriptions, the app could prove to be highly beneficial for the blind users as well as describers.

### **8.2 Future Work**

- Make UI design changes after user feedback.
- Improve Pass 2 to make the website live.
- Inline and extended descriptions alignment with the Youtube video.
- Matching with speed play of video with description.
- Have to set a pace to play video as there might be some hearing problems for some viewers.
- For Caption Rating Game, more users like to play games over the phone need to implement a phone version for the game.

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